

Shocks reshape human capital

Three rigorous economics papers · two current applied-development studies and one health-economics classic. The following article pages are edited reading excerpts, not complete replacements for the originals.

LITERATURE PULSE

The three papers show that shocks to health and the environment alter behavior and human capital through channels that standard outcome measures can miss. In Indonesia, deadly disasters leave a long shadow on willingness to take risk. In rural Pakistan, sustained particulate exposure slows learning, mostly through mechanisms beyond measured absence. The Malawi classic shows the reverse possibility: correcting nutrition knowledge changes diets, work, and child growth even without giving households food or cash. **Literature trend:** applied development economics is increasingly tracing mechanisms across domains—beliefs into risk choices, pollution into cognition, and information into household production. Fixed effects, a crop-fire instrument, and a cluster RCT each strengthen causal claims, but each answers a different counterfactual and carries its own identifying assumptions.

01 CURRENT / DISASTERS AND BEHAVIOR · STARTS ON PAGE 2 · World Bank Economic Review · September 2026 · peer reviewed · CC BY 4.0

The Heterogeneous Impacts of Natural Disasters on Risk Preferences in Indonesia

Do disasters change the willingness to take economic risks? The paper links two waves of the Indonesian Family Life Survey to district disaster records and estimates individual fixed-effects models for 7,947 adults. Cumulative disaster mortality raises measured risk aversion for at least seven years, with threshold effects and especially strong responses to deadly earthquakes. The design removes stable individual and district differences, but two survey waves cannot rule out all time-varying local shocks, and the outcome is a hypothetical lottery. The result matters because disaster damage may persist through later savings, migration, and technology-adoption choices.

02 CURRENT / ENVIRONMENT AND EDUCATION · STARTS ON PAGE 17 · World Bank Policy Research Working Paper 11463 · September 2026 · CC BY 3.0 IGO

Air Pollution and Student Learning in Pakistan

How much does dirty air slow learning, rather than merely depress performance on a bad test day? The authors combine longitudinal primary-school test data from rural Punjab with satellite PM2.5 and instrument sustained exposure using upwind crop-residue fires. A 1 microgram-per-cubic-meter increase in annual PM2.5 lowers the share of difficult questions answered correctly by about 2.8 percentage points in the preferred specification; absences explain less than one-tenth of the estimated loss. The instrument is relevant and extensively stress-tested, but the exclusion restriction could fail if upwind fires affect schools through channels other than pollution.

03 CLASSIC / HEALTH AND HOUSEHOLD BEHAVIOR · STARTS ON PAGE 32 · Journal of Development Economics 122 · 2016 · peer reviewed classic · CC BY 4.0

Nutrition, Information and Household Behavior: Experimental Evidence from Malawi

Can information alone improve child health when households must finance the response themselves? A cluster-randomized program in rural Malawi sent trained local women to make repeated home visits about infant nutrition. The intervention changed diets toward protein, fruit, and vegetables, increased household food consumption and male labor supply, and raised height-for-age by 0.27 standard deviations among children older than six months. Cluster counts limit precision and some component results are exploratory, but the paper's distinctive contribution is to trace health knowledge into household production, labor, and consumption choices rather than stopping at knowledge scores.

Editorial note: references and appendices were omitted where possible; no included article contributes more than 15 text pages. Full citations and source links remain in the catalog. AI-written synopses were checked against the included source pages.

The Heterogeneous Impacts of Natural Disasters on Risk Preferences in Indonesia

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Abstract

Many economic decisions are influenced by individual risk preferences, and new evidence challenges the immutability of these preferences, suggesting life experiences can shape our attitudes towards risk and uncertainty. This study explores the impact of disasters on individual risk attitudes using longitudinal data from Indonesia, focusing on the heterogeneity of disasters by type, severity, and timing. Results show risk aversion increases with increasing disaster mortality for at least seven years, and high-mortality disasters, namely earthquakes, are more salient to individuals than higher-frequency, lower-mortality disasters. These outcomes shed light on how disaster survivors in a developing country respond to and internalize these shocks and are informative to policymakers in addressing the increasing threat of severe disasters due to climate change and growing population density.

JEL classification D12, D81, O12, Q54

Keywords risk preference, natural disasters, Indonesia

1. Introduction

It is well documented that risk and uncertainty play a significant role in determining economic choices,¹ and research now suggests that these preferences are not immutable throughout an individual's life, particularly in the face of significant events. However, the evidence on whether and to what extent shocks, including natural disasters, financial crises, and conflict, impact individual risk preferences is inconclusive. This could have particularly important consequences for individuals in developing countries given that poorer households display higher levels of risk aversion as a result of ineffective institutions, lack of educational opportunities, poor health, exposure to violence and crime, and other economic challenges (Guiso and Paiella 2008; Haushofer and Fehr 2014; Rosenzweig and Binswanger 1993). This higher level of risk aversion may lead to underinvestment in risky but potentially rewarding behavior and activities, and therefore have negative economic impacts, and shocks may compound the consequences of credit constraints and uninsurable risks that exist for these individuals (Haushofer and Fehr 2014).²

¹ Including savings (Dohmen et al. 2011; Rosenzweig and Stark 1989; Lawrance 1991), investment in education and health (Dasgupta et al. 2015; Shaw 1996; Strauss and Thomas 1998), as well as technology adoption (Liu 2013; Feder 1980; Kebede 1992).

² For example, because of the environments they live in, the poor may be less willing to adopt new technologies (Liu 2013), be less entrepreneurial (Kihlstrom and Laffont 1979; Guiso and Paiella 2008), have short-sighted views on savings and

Given the traumatic nature and devastating human costs of natural disasters, it is plausible that they would impact an individual's view of risk and uncertainty. However, it is not immediately clear whether natural disasters would make individuals more risk averse or more risk tolerant. Indonesia is located on the Pacific Ring of Fire, and is frequently plagued by a variety of major disaster events including earthquakes, tsunamis, volcanic eruptions, wildfires, floods, landslides, and tornadoes. Additionally, there is substantial regional heterogeneity of economic and social development, which makes Indonesia an ideal research setting for exploring the varied ways in which natural disasters may influence risk attitudes. Using data from Indonesia, I investigate whether the intensity and severity of natural disasters over time impact individual risk attitudes. I combine data on risk attitudes from two waves of the Indonesian Family Life Survey (IFLS), a longitudinal household survey, with national natural disaster statistics reported by Indonesia's National Disaster Management Authority (BNPB) at the district level to assess whether increased mortality and destruction from disasters changes attitudes toward risk measured by a hypothetical lottery.

Currently, the literature on risk and disasters is not only conflicted on whether natural disaster shocks change individual risk attitudes, but also the direction of the changes and the transience of the change.³ Many studies lack the longitudinal data necessary to identify changes within individuals over time, or focus on a single disaster event, typically one of the most severe disasters in the country's history, rather than the perpetuation of disasters or the analysis of different types of disasters, which could have differential impacts. While [Hanaoka, Shigeoka, and Watanabe \(2018\)](#) are able to use panel data to evaluate the before and after changes in risk attitudes, they focus on a single event, the largest earthquake in Japan's history, in a developed country context. While their results may be valid for the developed country context, they may not hold true in a developing country such as Indonesia or for other types of disaster. This paper contributes to the literature by using panel data from a developing country in order to account for time-invariant unobservables, using unique disaster data that allow me to explore the heterogeneous impacts of disasters on risk.

This paper contributes to the literature on shocks and risk attitudes in the following important ways. First, it is the first to use panel data from a developing country to examine the impact of disasters on risk attitudes, accounting for time-invariant unobservables in order to address an important source of selection bias. I can therefore make stronger causal claims between disasters and risk relying on weaker stochastic assumptions compared to the current literature. Second, the unique aspects of my disaster data allow me to explore the heterogeneous impacts of disasters on risk. Specifically, I find that the measure of disaster severity, the type of disaster (earthquake, flood, etc.), and timing of disasters are important for explaining risk attitudes. The results from this paper can help inform policy surrounding disaster risk reduction as changes in risk attitudes will likely affect savings, consumption, farming, and livestock, as well as other important economic decisions made at the individual level.

I find that individuals become more risk averse as the cumulative district mortality rate and district destruction from disasters increase. The data point to the fact that the accumulation of experiences during a specific time frame matters more than any one individual event or year. While disasters are also likely to affect economic and health outcomes at critical periods in life, such as childhood and adolescence, the IFLS is limited to respondents aged 15 and above, and still answers a question about whether experiences in adulthood can affect our risk attitudes. This research is novel in that it does not study once-in-a-lifetime shocks such as the Japanese Earthquake or even the 2004 Indonesian Tsunami, but rather the more frequent lived experience of multiple disasters over time.

Results indicate that individuals become more risk averse as district mortality increases for at least seven years, with the first two years having less precise estimates. The data prohibit direct evaluation

financial investment ([Guiso and Paiella 2008](#)), and may underinvest in health and education, thus making it difficult to escape poverty traps.

³ [Cameron and Shah \(2015\)](#), [Cassar, Healy, and von Kessler \(2017\)](#), [Brown et al. \(2018\)](#), and [Beine et al. \(2020\)](#) find that natural disaster shocks increase risk aversion in developing countries. In contrast, studies by [Hanaoka, Shigeoka, and Watanabe \(2018\)](#), [Eckel, El-Gamal, and Wilson \(2009\)](#), [Bchir and Willinger \(2013\)](#), and [Page, Savage, and Torgler \(2014\)](#) find the opposite result: natural disasters cause individuals to become more risk tolerant or have no impact on risk preferences. [Cameron and Shah \(2015\)](#) provide detailed background on why both an increase in risk aversion or an increase in risk tolerance could be plausible, because of both increased background risk ([Guiso and Paiella 2008](#)) and prospect theory ([Kahneman and Tversky 1979](#)), respectively.

of disasters on risk preference past seven years due to overlapping survey rounds, but we have some evidence that the effect is reversed or fades in the long run. I also find that the severity of the disaster matters, and there is evidence of threshold effects. Below certain thresholds, disasters do not significantly impact risk aversion, while disasters significantly increase risk aversion above the threshold. I additionally find evidence that the changes in risk attitude may be driven by specific types of disaster, namely earthquakes (which often result in a high number of deaths and are more unpredictable than other types of disaster such as floods). Mortality appears to be the most salient of severity measures in its impact on risk attitudes among respondents, and may indicate that the death toll matters more than destruction or other measures such as evacuations. This is consistent with the psychology literature linking major natural disasters to psychological impacts such as post-traumatic stress disorder and depression, typically caused or exacerbated by the traumatic experience of injury and death, specifically the fear of death or death of a loved one, as well as personal property loss (Briere and Scott 2015).

2. Data and background

2.1. IFLS data

The Indonesian Family Life Survey is a household survey conducted in Indonesia that started in 1993, and because of the extensive effort to find and survey individuals that had moved or migrated, the IFLS was able to reinterview close to 90 percent of the original households, including split-off households. Since the initial survey wave in 1993, there have been five waves of surveys, the most recent of which was completed in 2015. The survey includes data on a variety of socioeconomic and demographic indicators at the individual, household, and community levels including age, ethnicity, religion, migration history, household expenditures, and availability of facilities in the community. This paper primarily uses the individual data from waves 4 (2007/8) and 5 (2014/15), as risk questions were only introduced in wave 4.

Beginning in fourth wave of the IFLS, a section on risk and time preferences was added to the questionnaire. In the risk section, hypothetical lottery choices are presented to all household members above the age of 15. The respondent is told, "Suppose you are offered two ways to earn some money." The respondent is then presented a scenario that provides two options, one where the payoff is certain, and one where the payoffs are uncertain. The uncertain option includes two possible payoffs with equal probability.⁴ There were two different sets of questions, one where the return was always positive (Game A) and one where there were possible zero outcomes or losses (Game B). See table 1 for the specific scenarios that were presented to the individuals.⁵ From the respondent's decision between the safe choice (certain payment) versus the risky choice (the gamble between a higher and lower amount than the certain choice with equal probability), I can estimate the individual's attitude towards risk. This analysis will focus on the risk preferences elicited from Game A, as the majority of respondents fall into the most risk averse category with Game B, and there may be differing interpretations of results when considering losses.

The responses to these questions put individuals on a risk scale from 1 to 4, where 1 is always choosing the certain option, and 4 is always choosing the gamble (excluding the Q1 response which is meant to be a logic check). An illustration of how the 1 to 4 scale is composed is shown in fig. 1.

A total of 8,990 individuals have risk data for both wave 4 and wave 5 of the IFLS. These individuals form the basis of my analysis as I am able to measure the change in risk over the two waves.⁶ I use three

⁴ While in other experiments and surveys this question is typically phrased as a gamble between two choices, due to the substantial Muslim population in Indonesia among whom gambling is prohibited, it is phrased in a way that the respondent would earn money with the option of becoming "lucky" and earning extra if the uncertain option is chosen.

⁵ If individuals choose the certain choice in Q1 they are asked again if they would choose the certain choice (there is only an upside for choosing the gamble in this scenario). If they continue to choose the certain choice, despite logic, the respondents are not asked any more questions. It is plausible that these respondents are so averse to uncertainty that they would choose the certain choice even when it does not make sense. Following Brown et al. (2019), I consider alternative specifications where these individuals are included as "gamble averse," with similar results to the main analysis which exclude these respondents.

⁶ There are a variety of reasons that individuals do not have data in both waves of the survey, such as dying between waves, refusing to respond, not being home during the interview, and being added to the survey in wave 5. There are also additional individuals who qualify as "gamble averse" in at least one wave and are not included in the 8,990.

(2015), by estimating risk aversion parameters for each individual in the data assuming Constant Elasticity of Substitution (CES) utility: $U(c) = c^{1-\theta}/(1-\theta)$. By defining utility over winnings from the risk experiment, I am able to calculate CRRA intervals for the four categories of risk aversion (shown in the CRRA line in [fig. 1](#)). Summary statistics by risk category can be found in the supplementary online appendix ([table S1.1](#), available with this article at the *World Bank Economic Review* website), which most importantly shows that males tend to make riskier lottery choices, consistent with the current literature ([Eckel, El-Gamal, and Wilson 2009](#)).⁷ If the risk measures from the IFLS lottery choices are a true representation of risk attitudes, they should be able to predict risky behavior of individuals. Before I examine the impacts of disasters on risk attitudes, I first validate the measures of risk obtained from the hypothetical lottery. I use IFLS data on risky behavior and analyze whether this behavior is correlated with the individual's risk attitude from the lottery choices. Following [Jakiela and Ozier \(2019\)](#), [Hanaoka, Shigeoka, and Watanabe \(2018\)](#), and [Brown et al. \(2019\)](#), I use a probit model to evaluate whether risk choices influence the likelihood of migrating for work, being self-employed, or smoking.⁸ Owning your own business and seeking out job opportunities away from home are both behaviors that may result in higher returns in the long run but are risky in the short run, and smoking is generally considered a health risk and as such we would expect individuals with a higher risk tolerance to be more likely to smoke. The results of this analysis are shown in appendix [table A2](#) and indicate that risk category based on lottery choices predicts observed risky behaviors.

Disaster Context and Data

Sitting on the Pacific Ring of Fire, Indonesia is exposed on many fronts to many types of disasters. There are earthquake faults throughout the country, and 129 volcanoes lie along its southern coast. In 2018, the Sulawesi Earthquake was the deadliest disaster globally, and the 2004 Indian Ocean Tsunami was one of the deadliest disasters in recorded history. The BNPB, National Indonesian Statistics Agency (BPS), and the UN Population Fund (UNFPA) estimate that 97 percent of Indonesian's population lives in areas exposed to disasters ([Purwo Nugroho, Ritonga, and Anggraini 2015](#)). Disasters in the data include floods, earthquakes, landslides, tsunamis, volcanic eruptions, wildfires, storms, and tornadoes.

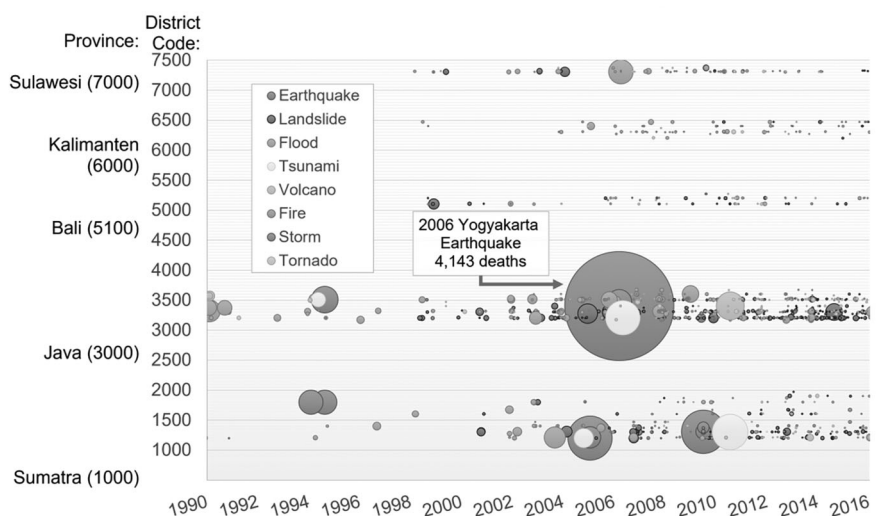
In 2007, following the historic 2004 Indian Ocean Tsunami, the Indonesian government passed legislation to strengthen disaster management and disaster relief in the country, including Law 24/2007 which established the National Agency for Disaster Management (BNPB) and called for new disaster risk reduction (DRR) plans for Indonesia. In 2008, the BNPB created a national natural disaster database called DIBI (Indonesian Disaster Data and Information) which took over from the National Disaster Management Coordinating Board (previously established in 1979).

Disaster statistics are reported regionally to the agency, which verifies the data and has published it publicly online. The database is organized by each individual disaster and includes basic information about the disaster, including type (earthquake, flood, terrorism), the date of the disaster, the districts that were affected, as well as several measures of the severity of the disaster. These severity measures include the number of people who died, were affected, evacuated, and wounded by the disaster, as well as the number of houses that were lightly, moderately, and severely damaged by the disaster and the number of worship, health, and education facilities that were damaged by the disaster. I focus specifically on natural disasters as they are plausibly more exogenous than other disasters such as conflict and epidemics, as the exposure to these types of shocks often depends on human behavior that might be correlated with risk preference. The dataset starts in 1815, but measurements and tracking of these numbers have likely gotten more intensive and accurate in recent years, especially as there has been a heightened focus on DRR by the Indonesian government since 2007.

There are frequent disasters that occur on a small scale, killing anywhere from 0 to 10 people, and are relatively minor events, such as floods and landslides. On the other extreme, there are high-impact events, such as earthquakes and tsunamis, that are less frequent but can kill thousands of people per event. [Figure 2](#) displays a timeline of disaster events (that apply specifically to the districts that appear in the sample) where the size and frequency of different types of disasters can be observed. The y-axis

⁷ For a full review of the literature on gender and risk see [Croson and Gneezy \(2009\)](#).

⁸ Only males are included in the probit for migrating for work.

Figure 2 Indonesian natural disaster events: *number of people killed by disaster (1990–2015)*

Source: Author's illustration of geographical dispersion and severity of natural disasters in Indonesia using data from the Indonesian Disaster Data and Information (DIBI) database run by the National Agency for Disaster Management (BNPB) in Indonesia.

Note: Disaster events in Indonesia by disaster severity, type and district over time. The size of the bubble represents the number of people killed by disaster. The y-axis plots the geographic location of districts by district code. The first two digits indicate the province, the second two digits represent the district. District codes are numerically ordered from west to east for the Southern Islands (Sumatra to Nusa Tenggara) and then again west to east for the Northern Islands (Kalimantan to Papua). Only IFLS districts are shown in the figure.

plots the four digit district code and is meant to display the geographic variation of the disasters. The largest of these events was the 2006 Yogyakarta Earthquake that killed thousands of individuals.⁹ For summary statistics and other details on district disasters, see the appendix.

3. Empirical specification

Cross-sectional and single-event studies are unable to capture variation over time and the heterogeneous impact of different disasters by severity and type. It is unlikely that all disasters impact individuals in the same way. This paper shows that the results are more nuanced than a yes or no answer to whether an individual has experienced a natural disaster. In fact, both time and severity matter greatly in determining changes in risk attitudes following disasters. Because the IFLS follows the same individuals over time, I am able to address concerns about unobserved heterogeneity using individual fixed effects. Even when controlling for some demographic characteristics, there may be individual unobservables as well as time effects that are formative to an individual's risk preference. For instance, it is possible that individuals with different risk preferences are choosing to live in areas that are more or less exposed to natural disasters. A cross-sectional approach will not capture the fact that more risk-averse individuals self-select into areas that experience fewer disasters. If individuals who are more risk averse choose to live in low-disaster regions, they are less likely to be exposed to disasters. In this case, the unobserved residential sorting is correlated with both disaster exposure and risk preference, and comparing risk preferences of those in disaster-exposed areas to non-exposed areas may simply be evaluating pre-existing differences between the two groups. Additionally, if risk-tolerant individuals move to unaffected areas following the disaster, researchers may falsely conclude that disasters induced an increase in risk aversion, when in fact the disaster simply induced less risk-averse people to move away and be excluded from the sample. If the comparison group then includes the less risk-averse individuals that moved into the non-affected area, the average level of risk aversion would be lower, confounding the results. In

⁹ The 2004 Indian Ocean Tsunami was the country's deadliest disaster to date, but the tsunami hit the Banda Aceh district, which is not included in the IFLS sample.

other words, fixed effects allow us to account for any of the district-level time-invariant unobservables that might be related to disaster exposure. I also use a population-adjusted measure for all district disaster data, as larger, more populated districts will be more likely to experience higher levels of death and destruction from disasters. A further study of district trends appears in the Robustness section.

Due to the panel nature of the IFLS and disaster data, I can follow the same individuals over time and I can therefore employ an individual fixed effects framework to evaluate the impact of natural disasters on risk attitudes accounting for the time-invariant unobserved individual heterogeneity. This strengthens the case for drawing causal claims between disasters and risk, relying on weaker stochastic assumptions compared to the current literature.¹⁰

Natural disasters are a plausibly exogenous shock. While there have been substantial advances in early warning systems and evacuation procedures, experts are still not able to tell exactly when or where a disaster will hit. Early warning systems in Indonesia were completed in 2008, so they are fairly new to the country, and are not perfect. For instance, the system failed in the 2018 Sulawesi Earthquake due to destruction of cell phone towers (people were unable to receive evacuation text alerts), a non-operational buoy network from lack of maintenance and vandalism, and lack of observation equipment in the area. Because of this, residents were not prepared for the earthquake and subsequent tsunami that hit the shores of Palu. While there are certain areas that are more prone to certain types of disasters (e.g., close proximity to fault lines and volcanoes, peat forests prone to wildfires, and flood-prone areas during monsoon), the timing and severity of disasters cannot be known by the individuals living in these areas. As mentioned above, the fixed effects remove the disaster hazard from the estimation, which means that the unpredictability relies on the timing and severity of the shocks. Even if one lives in a flood-prone or earthquake-prone area, experts agree that it is impossible to predict exactly and how hard a natural disaster will hit at any given time. As such, disasters can be treated as a random shock, given I control for selection caused by decisions made to live in a particular location.

For comparison purposes, it is useful to look first at the results from the cross-sectional results from both waves of the IFLS, which is the focus of the current literature in the developing country context. I run separate regressions for waves 4 and 5 of the IFLS using [equation \(1\)](#) below, controlling for individual demographic characteristics.¹¹ [Equation \(1\)](#) assumes that there is zero correlation between experiencing a disaster and the time-invariant as well as time-varying unobservables, as the two are indistinguishable in a cross-sectional model. I then remove individual-specific time-invariant unobservables and IFLS wave characteristics using a fixed effects specification, shown in [equation \(2\)](#). This specification relies on weaker stochastic assumptions than [equation \(1\)](#) as it only assumes zero correlation between the time-varying unobservables and the disaster term.

$$\text{Risk}_{ij} = \alpha_0 + \beta_1 D_j + \sum_{k \in K} \beta_k X_{ki} + \epsilon_{ij}, \quad (1)$$

$$\text{Risk}_{ijt} = \alpha_0 + \beta_1 D_{jt} + \sum_{k \in K} \beta_k X_{kit} + \delta_i + \sigma_t + u_{ijt}. \quad (2)$$

In the specifications above, Risk_{ijt} refers to the risk category of individual i , in district j , at survey wave t . This ranges from 1 (most risk tolerant) to 4 (most risk averse). I also run additional specifications using a risk-averse dummy and the coefficient of relative risk aversion as the risk preference parameter. Cumulative district mortality rates from disasters is represented by D_{jt} , a continuous variable that is measured from the prior 1 to 7 years. I additionally include other measures to capture disaster severity, including house destroyed and economic damages and time periods. The variable X_{ki} is a vector of demographic controls for individual i , while X_{kit} represents the vector of time-varying characteristics, such as marital status and age. Additionally, δ_i and σ_t represent individual and wave fixed effects, respectively. All standard errors are clustered at the district level. The main specification in [equation \(2\)](#) shows the individual impact of an increase in district mortality from disaster on risk attitudes, where β_1 represents the increase (or decrease if negative) in risk-aversion category given additional exposure of 1 death per 1,000 people in the district.

¹⁰ For additional evidence that a fixed effects model is preferred to a random effects model, I use a cluster-robust Hausman test ([Wooldridge 2002](#)) to provide support for using a fixed effects model. The null hypothesis of the Wald test is rejected, implying that the fixed effects model is preferred (p -value < 0.01).

¹¹ Individual controls include age, gender, ethnicity, religion, marital status, and a dummy for living in an urban area.

4. Results

4.1. Intent-to-treat effects of disasters on risk aversion

Results show that individuals who are exposed to increasing district mortality from disasters become significantly more risk averse for a sustained period of time. There are multiple ways to measure whether an individual has been exposed to a natural disaster. For the main measure of severity I use the district mortality rate, similar to [Gennaro, Bhagwat, and Rau \(2017\)](#), who use the county mortality rate of a variety of disasters to evaluate their impact on Chief Executive Officer (CEO) risk behavior.¹² Recall that the measure of disaster used in the main results is cumulative mortality, adjusted for district population. I find that using individual disaster counts or disaster deaths during a particular year yields inconsistent results, and the true impact of disasters is revealed when the experiences are accumulated over time. When individual disasters or years are evaluated, there is no significant change in risk preferences, likely because individual years are correlated with each other, or that separately they do not have a large enough impact, and it is the cumulative effect that matters most. In fact, [Beine et al. \(2020\)](#) find that a second earthquake in Albania has a cumulative effect on risk aversion, equal in magnitude to the impact of the first earthquake, showing that experiencing multiple disasters can compound changes in risk preference. The cumulative impact of disasters on risk can only be evaluated for the prior seven years in this manner, as past that time the survey rounds overlap and the comparison using fixed effects does not allow for proper interpretation.

[Table 2](#) shows the impact of an additional district death per 1,000 people in the district from disaster on an individual's risk category (1–4) over multiple time periods.¹³ The table shows that if we only consider disaster mortality over the past year, the coefficient is high but imprecise, likely a function of the low district mortality over the past year. If I instead consider the cumulative mortality over, say, the last five years, which will include the first year plus the previous four years of district deaths, then the impact is positive and significant.¹⁴ [Table 3](#) shows the impact of adding each additional year of disaster deaths to the prior cumulative deaths to see how this relationship evolves.

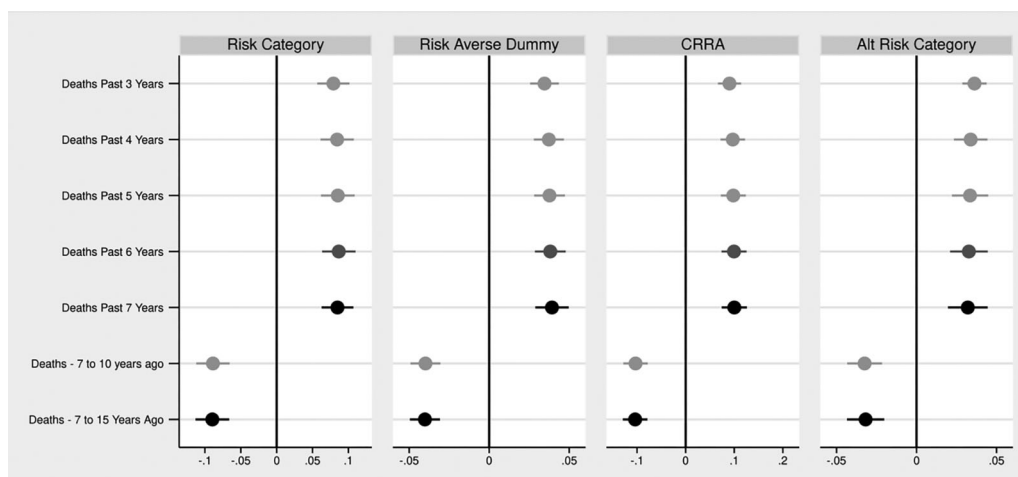
I also plot these coefficients using the cumulative mortality from 3 to 7 years, as seen in [fig. 3](#). The results from 1 and 2 years are suppressed because of their out-sized coefficient sizes and large confidence intervals which distort the estimates in other years.¹⁵ An increase in district deaths measured over a 3- to 7-year time frame causes a statistically significant increase in risk aversion measured by risk category. Overall, once mortality has accumulated into the third year in the past, risk aversion changes significantly. This is true whether you look at cumulative mortality over 3 years or up to 7 years. Unfortunately, because of the timing of the IFLS surveys, the fixed effects framework does not allow for longer-term analysis in this manner because of overlapping time frames. As an alternative, although it is not directly comparable, I accumulate deaths past 7 years up to 10 or 15 years. This would not take into account the recent disaster experience, but does give some insight into what the longer-term impact on risk would be. See [table 4](#) and the last two rows of [fig. 3](#) which show that longer-term deaths induce an increase in risk tolerance. However, one must be somewhat skeptical with interpretation, as this is close to the equal and opposite effect of the most recent 7 years, and may together signal a fading effect by canceling out rather than an increase in risk tolerance.

¹² While the IFLS has a simple question posed to households about whether they experienced a natural disaster in the last five years, to which the respondent can answer yes or no, there is virtually no information on severity and it only captures the last five years. Additionally, a disaster could mean different things to different people. In order to avoid possible bias of the self-reported measure of disaster in the IFLS and gain more detail on disaster severity and exposure, I use national disaster statistics reported at the district level in Indonesia as my main measure for disaster.

¹³ See appendix A4 for the cross-sectional comparison using [equation \(1\)](#).

¹⁴ As the main measure of risk preference is a categorical measure, it may be argued that a linear estimation is not appropriate. I alternatively estimate [equation \(2\)](#) using an ordered logit with individual fixed effects. The results are similar in direction and significance to the main results. Marginal effects can only be measured at the average, and results show that an increase of 1 death per 1,000 people in the district leads to a 3-percent increased likelihood of falling into the most risk-averse category, and a 3-percent decreased probability of falling into the least risk averse category. It should be noted that this estimation excludes those individuals who do not have a change in risk category between rounds.

¹⁵ While the coefficients appear large, the 1- and 2-year mortality rates from disaster are small, and when this is taken into consideration, the coefficients are in line with the results for other time frames.

Figure 3 Coefficient estimates for risk on disaster deaths, by risk type for all disasters

Source: Author's estimation based on data from waves 4 and 5 of the Indonesian Family Life Survey (IFLS) and natural disaster data from the Indonesian Disaster Data and Information (DIBI) database run by the National Agency for Disaster Management (BNPB) in Indonesia. Note: Coefficient estimates for separate regressions using district deaths from disaster as primary independent variable and confidence intervals. Standard errors are clustered at the district level. All regressions use individual and survey wave fixed effects and control for demographic characteristics. Deaths past 7 years are measured from 7 years ago up to 15 years ago, to avoid overlap in the data.

Table 4. Risk category on total district deaths from all disasters *adjusted for district population (per 1,000)*.

	Risk category		
	(1)	(2)	(3)
7-year mortality	0.0848*** (0.0112)	–	–
Mortality from 7–10 years ago	–	–0.089*** (0.012)	–
Mortality from 7–15 years ago	–	–	–0.090*** (0.012)
Avg period mortality rate	0.102	0.0843	0.089
Observations	15,894	15,894	15,894
Number of respondents	7,947	7,947	7,947
Adjusted R-squared	0.0210	0.0212	0.0213

Source: Author's estimation based on data from waves 4 and 5 of the Indonesian Family Life Survey (IFLS) and natural disaster data from the Indonesian Disaster Data and Information (DIBI) database run by the National Agency for Disaster Management (BNPB) in Indonesia.

Note: Outcome variable measure on a scale from 1 to 4, where 1 is most risk tolerant and 4 is most risk averse. All regressions include individual fixed effects, round fixed effects, and demographic controls. Mortality rate is per 1,000 people in district. Standard errors in parentheses, clustered at the district level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

threshold) and it results in imprecise estimates, or that there is an immediate emotional response to the disaster which conflicts with the change in risk in the longer term. Eckel, El-Gamal, and Wilson (2009) find that Katrina evacuees are more risk-loving in the short term than the longer term and compared to a comparable control sample, and that this is explained by negative-emotion variables. Additionally, most of the studies done in this literature focus on large-scale natural disasters (e.g., the Great Japanese Earthquake, Cyclone Evan, the 2004 Indian Ocean Tsunami), results of which may be reflective of the severity of these events. It is possible that immediately following disasters, individuals are either in shock or expect aid to come, and this period of limbo distorts the risk measure. To investigate this, I

Table 5. Impact of severe disasters.

	Risk category		
	(1)	(2)	(3)
1-year mortality	8.806*** (1.528)	–	–
5-year mortality	–	0.085*** (0.011)	–
7-year mortality	–	–	0.084*** (0.011)
Avg mortality rate	0.001	0.085	0.093
Number of respondents	7,947	7,947	7,947
Observations	15,894	15,894	15,894

Source: Author's estimation based on data from waves 4 and 5 of the Indonesian Family Life Survey (IFLS) and natural disaster data from the Indonesian Disaster Data and Information (DIBI) database run by the National Agency for Disaster Management (BNPB) in Indonesia.

Note: Outcome variable measure on a scale from 1 to 4, where 1 is most risk tolerant and 4 is most risk averse. All regressions include individual fixed effects, wave fixed effects, and demographic controls. Severe disasters defined as 10 or more people dying or affecting 100 people or more. Mortality rate is per 1,000 people in district. Standard errors in parentheses, clustered at the district level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

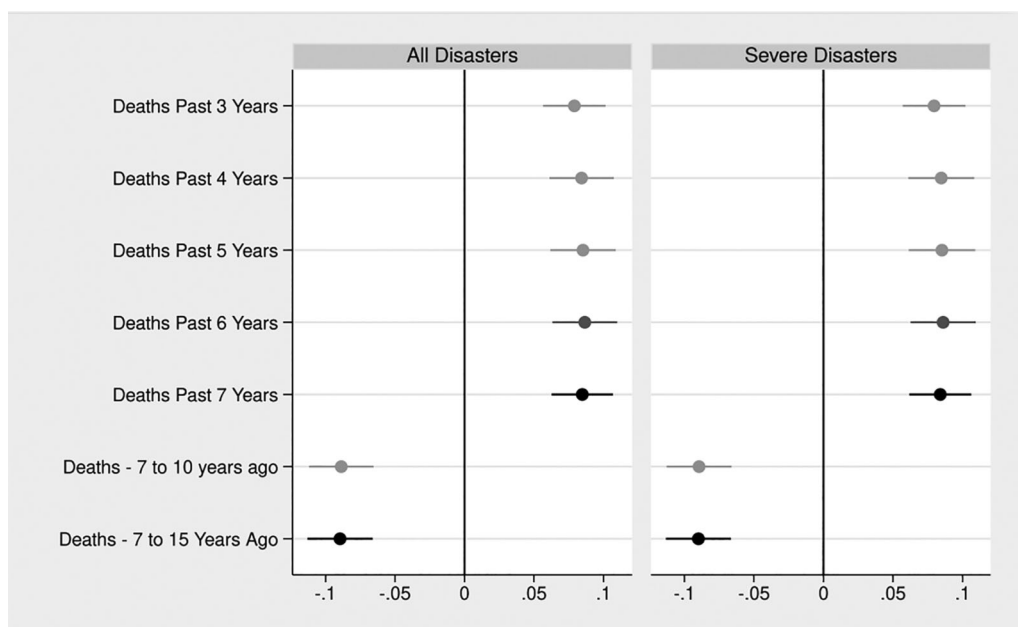
first reestimate [equation \(2\)](#) to only include disaster deaths for severe disasters as defined by EM-DAT.¹⁶ [Table 5](#) shows that increases in deaths from severe disasters have a positive and significant impact on risk even in the short run.¹⁷ The results for disasters measured over a longer time frame mirror the direction and significance displayed in the main specification using all disasters. [Figure 4](#) shows that for severe disasters, the initial increase in risk aversion is higher. There is very little evidence on the long-term impacts of natural disasters, and it is somewhat confounding why individuals would be less risk averse many years after the disaster occurred. However, [Frankenberg, Sumantri, and Thomas \(2020\)](#) do find that mortality risks increase 10 years after the 2004 Indian Ocean Tsunami and [Gennaro, Bhagwat, and Rau \(2017\)](#) find that moderate disaster fatality experiences in childhood lead to riskier decisions by CEOs later in life. It is possible that individuals face higher mortality risks in the long term following disasters, which impacts their perception of risk and uncertainty, or that making it through the disaster experience somehow desensitizes individuals to risk in the long term. This may also point to additional resilience among the most affected people, where surviving such a serious life shock means that you are able to bounce back to the point of becoming even more risk tolerant than before experiencing the disaster.

4.2. Threshold effects

Using a spline regression, I further test the theory that more fatal and destructive disasters are more salient to individuals and therefore more likely to change risk attitudes. I investigate whether the data is kinked, specifically whether a threshold exists, where for individuals who experience lower levels of district deaths (i.e., less severe disasters) there is a different risk response than those who experience disaster deaths above a certain threshold. [Equation \(3\)](#) shows the spline specification, where T is the

¹⁶ EM-DAT and the Red Cross only consider disasters severe enough to include in their data if at least 10 deaths or more than 100 people are affected by the disaster (<https://www.emdat.be>).

¹⁷ The coefficient on the one-year mortality rate seems impossibly large, but this simply reflects the steep slope of the regression. Given the one-year mortality rate is below 1, an increase in 1 death per 1,000 residents would not be possible. The coefficient is better interpreted if you consider an increase of 1 death per 10,000, which would lead to 0.8 increase in risk category.

Figure 4 Coefficient estimates for risk on disaster deaths, all disasters and severe disasters

Source: Author's estimation based on data from waves 4 and 5 of the Indonesian Family Life Survey (IFLS) and natural disaster data from the Indonesian Disaster Data and Information (DIBI) database run by the National Agency for Disaster Management (BNPB) in Indonesia. Note: Coefficient estimates for separate regressions using district deaths from disaster as primary independent variable and confidence intervals. Standard errors are clustered at the district level. All regressions use individual and survey wave fixed effects and control for demographic characteristics. Severe disasters are classified as those that have killed 10 or more people or affected more than 100 people.

threshold tested depending on the disaster measure (deaths, damages, houses destroyed):¹⁸

$$\text{Risk}_{ijt} = \alpha_0 + \alpha_1 I_b[D_{jt} < T]D_{jt} + \alpha_2 I_a[D_{jt} \geq T]D_{jt} + \sum_{k \in K} \alpha_k X_{kit} + \delta_i + \sigma_t + u_{ijt}. \quad (3)$$

I find that below the threshold of 7 deaths per 1 million individuals in the district there is no significant change in risk preference, and above the threshold, district disaster mortality has a positive and statistically significant effect on risk aversion; see [table 6](#). I also do alternative measures of the spline where economic damages and houses destroyed are used in place of deaths and a similar threshold effect exists above \$114,000 in damages and above 0.170 houses destroyed (both population adjusted).¹⁹ This indicates that individuals must be hit by a significant enough shock for it to impact their risk preference, small shocks below a certain threshold may not be salient enough to cause a shift. Interestingly, there is an increase in risk tolerance below the damages threshold, although the effect is small.²⁰ This result is consistent with [Gennaro, Bhagwat, and Rau \(2017\)](#) who find that CEOs who experienced disasters that were not hugely negatively consequential were more risk taking later in life. It could be true that individuals who experienced disasters but were lucky enough that their communities did not suffer drastically from an economic standpoint had a reaction that was in contrast to those who were not so lucky. Additionally, to distinguish between a frequency and severity effect, I run regressions with disaster counts, which have a negative and statistically insignificant impact on risk preference (individuals become more risk tolerant). This further indicates that it is large, infrequent disasters that are driving the main results. It makes sense that larger, more deadly and destructive disasters would be more

¹⁸ I use the following thresholds: 1 death per 100,000 people in the district, \$100,000 in damages, and 20 houses per 1,000 people in the district. The indicator I represents whether the disaster measure is below or above the threshold. Because the district deaths are measured per 1,000 people, the indicator equals 0.01 in the equation. Most districts have a population between 500,000 and 1 million people so this would equate to about 5–10 people dying in disasters over the last 5 years.

¹⁹ Spline cutoffs determined by sample median among those who experienced any deaths, damages, and houses destroyed, respectively.

²⁰ Maximum effect of $-1.10 * 0.114 = -0.125$ in risk category compared to the maximum effect above the threshold of 0.59 ($= 0.025 * 23.6$).

Table 6. Spline regressions.

	Risk category		
	Deaths (1) = 0.007	Damages (2) = 0.114	Houses (3) = 0.170
Spline cutoff:			
Below cutoff	−8.815 (13.12)	−1.096** (0.454)	0.040 (0.446)
Above cutoff	0.093*** (0.019)	0.025** (0.011)	0.003*** (0.001)
Maximum value	7.5	23.6	345.0
Observations	15,894	15,894	15,894
Number of respondents	7,947	7,947	7,947
Adjusted <i>R</i> -squared	0.021	0.021	0.021

Source: Author's estimation based on data from waves 4 and 5 of the Indonesian Family Life Survey (IFLS) and natural disaster data from the Indonesian Disaster Data and Information (DIBI) database run by the National Agency for Disaster Management (BNPB) in Indonesia.

Note: Outcome variable measure on a scale from 1 to 4, where 1 is most risk tolerant and 4 is most risk averse. Spline cutoffs determined by sample median among those who experienced any deaths, damages, and houses destroyed, respectively. Deaths measured as deaths per 1,000 residents in district from disasters in last five years. Damages measured in millions of US dollars from disasters in last five years per district, adjusted for population. Houses destroyed is measured as the number of severely damaged houses per 1,000 residents in district in the last five years. Robust standard errors in parentheses, clustered at the district level. Spline regressions include both individual and round fixed effects. Averages are calculated for individuals who experienced disasters above the threshold. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

salient to individuals than smaller more frequent disasters as it is likely the shock and devastation that causes a shift in preferences, rather than constant exposure to smaller threats that become part of daily life.

4.3. Heterogeneity: Severity measurements, disaster types, and subgroups

So far the main measurement used to evaluate the impact on risk attitudes is deaths from disaster. There are, however, other measurements of disaster severity that can be used to explore whether the results are consistent across measurements. Alternative measurements from the BPS database include evacuations, houses destroyed, and number of people affected. Additionally, EM-DAT, the International Disasters Database, reports both deaths and damages from disasters for each country. I find that damages and the number of houses destroyed broadly follow the patterns seen using disaster mortality, but the impacts are not as large in magnitude and the effects on risk from houses destroyed are quicker to fade. In the case of damages, the results are not statistically significant, but both deaths and damages follow a similar pattern over 15 years as the BNPB results (see [fig. 5](#)). Evacuations and the number of people affected seem to have little effect on individual risk attitudes, showing some short-run increases in risk tolerance, but virtually no effect afterward. A detailed discussion of these results, including EM-DAT regressions, houses destroyed, and evacuations, can be found in section S3 of the supplementary online appendix ([table S1.2–S1.4](#)).

From the variety of measurements for disaster and the different subsets of the sample that are tested, it is clear that the impact of disasters is not homogeneous. Death has a more significant influence on risk than other measures, such as evacuations and destruction from disasters. This may indicate that while risk attitudes change following the trauma of the natural disaster, there is evidence of resilience or a return to the status quo, after a certain amount of time has passed. There is evidence in the psychological literature that suggests that those who have longer-lasting psychological effects from natural disasters

Table 7. Impact of disasters on mental health.

	Standardized score for depression					
	Full sample (1)	Females (2)	Males (3)	Full sample (4)	Females (5)	Males (6)
1-year mortality	33.05* (17.07)	47.91* (27.41)	22.31* (13.28)	–	–	–
5-year mortality	–	–	–	0.17*** (0.04)	0.28*** (0.06)	0.08** (0.04)
Observations	15,894	7,736	8,158	15,894	7,736	8,158
Number of respondents	7,947	3,868	4,079	7,947	3,868	4,079
Mean mortality	0.002	0.002	0.002	0.092	0.089	0.094
Adjusted R-squared	0.164	0.163	0.166	0.163	0.162	0.166

Source: Author's estimation based on data from waves 4 and 5 of the Indonesian Family Life Survey (IFLS) and natural disaster data from the Indonesian Disaster Data and Information (DIBI) database run by the National Agency for Disaster Management (BNPB) in Indonesia.

Note: Depression measure is a standardized score based on 10 mental health questions asked in the IFLS. Standard errors in parentheses, clustered at the district level. All regressions include individual fixed effects, wave dummy, and demographic controls. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

and perceptions of risk. The IFLS includes information on mental health. The IFLS instructs that the 10 questions on mental health and emotions can be indexed into a depression score, where individuals are considered depressed if they are above a score of 10, which I have standardized for the sample. While only suggestive, [table 7](#) shows that there is a relationship between natural disaster mortality and an individual's mental health and propensity for depression at the one- and five-year time frame, which may indicate that it is through emotions that individuals display different risk preferences.

The second pathway through which disasters are inducing a change in risk aversion is through income, where disasters cause a loss in wealth and that loss in wealth drives individuals' risk preferences. The IFLS also has information on individual and household income. I run a similar regression to the mental health analysis to investigate whether risk attitudes may be impacted through an income channel. If this is the case, individuals may appear more risk averse because their risk is evaluated at a lower income level. In fact, [table 8](#) shows that district disaster mortality is generally negatively correlated with household income, and is significant for five-year mortality, when we see the most significant impacts on risk aversion. Individual income impacts are even more persistent and precisely estimated, and overall this points to the fact that loss of income is a consequence of disasters, which in turn could affect an individuals' risk aversion.

The third possibility, as cited by [Cassar, Healy, and von Kessler \(2017\)](#) and [Cameron and Shah \(2015\)](#), is that individuals update their expectations of future events occurring once they are hit by a disaster they did not expect. The likelihood of being struck by an earthquake or tsunami are low, but once an individual has already experienced this, they may be more inclined to believe it will happen again, thereby adding to the background risk an individual perceives. This change in expectations may make individuals perceive that the world is a riskier place because of their experience. [Cameron and Shah \(2015\)](#) do show some evidence of probability updating, but unfortunately none of the IFLS questions ask about future probabilities of events or expectations so it would be difficult to test for the sample. This result is also found by [Gibson and Mullins \(2020\)](#), who find that the decrease in property values resulting from Hurricane Sandy and updated Federal Emergency Management Agency (FEMA) maps are driven by belief-updating regarding flood risk in the area. They emphasize the importance of belief updating, especially as climate change is increasing the risk of disaster events.

Belief updating could have important implications for policymakers, as the demand for insurance may increase following disasters and there is evidence that the distribution of disaster losses may be fat tailed in nature. [Conte and Kelly \(2021\)](#) explore the implications of fat-tailed events, where disaster

Table 8. Impact of disasters on income.

	Log household income			Log individual income		
	(1)	(2)	(3)	(4)	(5)	(6)
1-year mortality	−2.213 (1.391)	–	–	7.5677 (9.019)	–	–
5-year mortality	–	−0.011* (0.007)	–	–	−0.203*** (0.040)	–
7-year mortality	–	–	−0.012 (0.008)	–	–	−0.204*** (0.041)
Observations	15,894	15,894	15,894	15,894	15,894	15,894
Number of respondents	7,947	7,947	7,947	7,947	7,947	7,947
Mean mortality	0.002	0.092	0.102	0.002	0.092	0.102

Source: Author's estimation based on data from waves 4 and 5 of the Indonesian Family Life Survey (IFLS) and natural disaster data from the Indonesian Disaster Data and Information (DIBI) database run by the National Agency for Disaster Management (BNPB) in Indonesia.

Note: Log of household income is annual and per capita; log of individual income is a sum of all annual labor income, non-labor income, and transfers. All regressions include individual fixed effects, wave dummy, and demographic controls. Standard errors in parentheses, clustered at the district level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

events measured by damages and mortality may not follow a normal distribution, and tail events may occur with higher frequency than expected. One property of these events is that the largest observations in the data are poor predictors for subsequent tail events. The fact that disasters are fat tailed may be driving the change in risk perceptions which requires belief-updating, since this would be unexpected if these losses were normally distributed.

6. Robustness

I examine the robustness of my results to a number of concerns. I test for the impact of district trends, selective attrition, analyze possible bias from endogenous migration, and compare results to regressions that include individuals who are “gamble averse.” Additionally, the relationship between risk and time preference is discussed in the supplementary online appendix (supplementary online appendix S5, tables S1.5 and S1.6, and fig. S1.2.) to validate the results of changes in individual risk aversion.

While the time-invariant characteristics of each district are removed via individual fixed effects and the time dummies capture any time trends, it is useful to test economic and development trends within the district are not impacting the main results. In order to account for different trends over time among the districts, I use data reported at the district level from the Indonesia Database for Policy and Economic Research (INDO-DAPOER), published by the World Bank from 1999 to 2015. For 2007 and 2014, I calculate the average annual growth rate for a variety of human and economic development variables over the previous five years, using round-specific windows (2002–2007 for Round 4; 2009–2014 for Round 5). I then control for these growth rates in equation (2) by adding a time-varying district control:

$$\text{Risk}_{ijt} = \alpha_0 + \beta_1 D_{jt} + \beta_2 R_{jt} + \sum_{k \in K} \beta_k X_{kit} + \delta_i + \sigma_t + u_{ijt}.$$

Results of this analysis are shown in table A7. The main coefficient on district disaster deaths remains positive and statistically significant across all specifications, providing reassurance that the main results are not driven by broader district economic trends. While most district trend controls are statistically insignificant, growth in per capita household expenditure and access to electricity are negatively associated with risk aversion, consistent with wealthier, more developed districts having better consumption-smoothing mechanisms that independently reduce risk aversion.

The main results may also be biased in the presence of selective attrition, endogenous migration, or the exclusion of what I call “gamble-averse” individuals from the analysis. I perform several checks to

ensure the robustness of the results over these domains, details of which can be found in the appendix. Overall, I find that 7,577 respondents were not interviewed or do not have risk information in wave 5 after being interviewed in wave 4, for reasons such as dying between rounds and having a proxy interview where the lottery questions were not asked. While these individuals do differ from those who were followed up in 2014 on some demographic differences, overall I do not find selective attrition based on district disasters or risk category in 2007. See appendix [table A8](#) and the Selective attrition subsection of the appendix for more information.

Selectivity from endogenous migration could be a concern if individuals moved from a district after being interviewed in 2007 and were not able to be traced in 2014 due to differences in disaster exposure and/or risk preference. Anyone who was not followed up after migration would be included in the attrition analysis described above, and out of the 2,000 respondents who migrated, only 4 migrated because of a natural disaster. Endogenous migration is a greater concern for cross-sectional studies where migration cannot be disentangled from selectivity on risk. For instance, if generally more risk-tolerant people move away from the district after a disaster and only the residents that remained in the district were interviewed, there would be selection bias in the sample. These concerns are largely addressed by following the same people over time and this further bolsters the importance of a longitudinal analysis. It is possible that the change in risk preferences would induce migration following disasters, where individuals become more risk averse and want to move away from the area after the disaster hits. If a disaster experience forced an individual to migrate and this in turn influenced the individual's risk preference, the impact of disasters on risk could be upward biased by not properly accounting for migration in the identification strategy. However, appendix table A9 shows that higher disaster mortality has no significant effect or results in less migration away from the district, addressing the concern that it is migration driving the increase in risk aversion, not the shock itself. Additionally, I test whether the results differ for those who have never moved from the district. While coefficient estimates are higher at the 5-year mortality and insignificant for 15-year mortality, the results are largely similar.

Finally, I consider respondents who answer the first lottery question, meant as a comprehension check, in a way that is not logical. They choose the certain choice even when the uncertain choice only has positive payoffs. This could be due to a lack of comprehension of what was being asked, or, as [Brown et al. \(2019\)](#) suggest, that these respondents would rather know with certainty the amount they will receive and are therefore "gamble averse." I analyze whether including these individuals in the analysis would affect the results, and find that they do not. A more comprehensive discussion of these individuals and the resulting analysis can be found in appendix [table A10](#).

7. Conclusion

Economists have long sought to understand the factors that influence our economic decision-making and lifetime earnings. There has been substantial debate over whether risk preferences are immutable throughout one's life or if experiences can shape these preferences. As [Schildberg-Hörisch \(2018\)](#) emphasizes, this question can largely be framed as an empirical one. This paper analyzes the impact of natural disasters on risk preferences in Indonesia, a country highly exposed to disasters.

This paper adds to the growing literature that attempts to measure the economic and psychological impacts of natural disasters and how individuals recover from the experience of a large shock. Governments are increasingly confronted with the challenges that climate change presents, especially for the most vulnerable populations, which have serious implications for development. Individual changes in risk aversion may inform policy and disaster recovery programs in a variety of ways, including insurance take-up, the importance of early warning systems, the willingness of individuals to change jobs or start their own businesses, investment in agricultural technology, as well as other important household decisions such as fertility and marriage. The differences in outcomes from a developed versus developing standpoint may be indicative of the importance of social safety nets and access to credit as mitigation mechanisms in low-income countries.

Using longitudinal data from a household survey and national disaster statistics in Indonesia, I evaluate how disasters impact risk attitudes, removing time-invariant individual heterogeneity by including individual fixed effects. The literature has produced mixed results on the impact of disasters on risk preferences due to primarily ex post disaster analysis and differences between developing and developed countries. I find that larger, more severe disasters affect risk attitudes more than extensive disasters with lower mortality. Death and destruction also have a more significant influence on risk than other measures, such as evacuations or the number of people affected by a disaster. This may indicate that while risk attitudes change following the trauma of the natural disaster, there is evidence of resilience or a return to the status quo, after a certain amount of time has passed. There is evidence in the psychological literature that suggests that those who have longer-lasting psychological effects from natural disasters stem from the traumatic experience of injury and death, specifically the fear of death or death of a loved one, as well as personal property loss. Intensive disasters which are less frequent and more destructive, such as earthquakes and tsunamis, drive the results, while extensive disasters, such as floods and landslides, which are more frequent and less severe, have a short-term impact that fades quickly. I also find suggestive evidence that both income loss and emotional responses may be the mechanism through which risk preferences are changing.

These results have important implications for policymakers, as increases in risk aversion can cause suboptimal levels of household investment and savings, and may make individuals less likely to open businesses or adopt new technologies. As climate change and population density continue to impact the human cost from disasters, governments should think about investing in early warning systems (as evacuations do not seem to cause any change in risk attitude) and other mitigation strategies which can reduce disaster mortality. Another important consideration for policymakers is the implications for disasters resulting in extreme levels damages and losses that occur with higher frequency than expected under a normal distribution, and would have consequences for insurance pricing and how individuals update their beliefs.

8. Data Availability

The IFLS data used in this article can be accessed from the RAND Corporation at <https://www.rand.org/labor/FLS/IFLS/download.html>. The BNPB disaster data can be accessed from the Indonesian National Disaster Management Authority at <https://dibi.bnpb.go.id>. The EM-DAT data can be accessed from the Centre for Research on the Epidemiology of Disasters at <https://www.emdat.be>. District-level data from INDO-DAPOER can be accessed from the World Bank Data Catalog at <https://datacatalog.worldbank.org>. A replication package producing all of the exhibits in the paper is available at the Harvard Dataverse, <https://doi.org/10.7910/DVN/JP9Z5U>.

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A.1. Summary Statistics

Table A1 shows summary statistics of the overall sample. On average, individuals fall between Category 2 and 3 on the four-point scale, and half of the sample falls in the most risk averse category. About half of the sample is male, the average age is 34 (recall the minimum age to respond to the risk questions is 15), 71% are married and live in households with an average of 4 people. Urban residents make up 55% of the sample and the majority (89%) are Muslim. The two largest ethnicities are Javanese (43%) and Sundanese (12%) and the average years of schooling is nine years. Average annual household income

Air Pollution and Student Learning in Pakistan

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Abstract

Ambient air pollution is a major health risk, but its implications for human capital formation are not well understood, particularly in low- and middle-income countries where exposure is high. This paper estimates the causal effect of sustained exposure to fine particulate matter on primary-school learning in rural Punjab, Pakistan. The paper combines longitudinal student achievement data from the Learning and Educational Achievements in Punjab Schools project with high-resolution satellite-based fine particulate matter estimates. To address endogeneity of pollution to local economic activity, the analysis exploits plausibly exogenous variation generated by open burning of agricultural residue interacted with wind direction, constructing school-year measures of upwind fire exposure as instruments for local fine particulate matter. The instrumental variable estimates show that higher school-year fine particulate matter significantly slows learning, with

larger effects in mathematics than in language subjects. The results are consistent with impacts operating during instructional months: pollution exposure during the school year predicts lower achievement, while exposure during vacation months has little explanatory power. The findings show statistically imprecise heterogeneity by age and gender, opposite to patterns found in comparable settings. The analysis is organized around a simple framework in which learning depends on attendance, cognition, effort, and instructional quality, each a function of exposure to poor air quality. Pollution increases student absences, but this attendance channel explains only a small portion of the total effect, implying that most of the learning loss is non-attendance-related and likely operates through the framework's other channels—direct effects on cognition, reduced effort, and degraded instructional quality.

This paper is a product of the Development Research Group, Development Economics, the Education and Skills Department, and the South Asia Region. It is part of a larger effort by the World Bank to provide open access to its research and make a contribution to development policy discussions around the world. Policy Research Working Papers are also posted on the Web at <http://www.worldbank.org/prwp>. The authors may be contacted at abehrer@worldbank.org, sasad1@worldbank.org, myanezpagans@worldbank.org, and mheger1@worldbankgroup.org. A verified reproducibility package for this paper is available at <http://reproducibility.worldbank.org>, click **here** for direct access.



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1 Introduction

In this paper, we ask how sustained exposure to ambient particulate pollution affects the pace at which children learn in one of the most polluted education systems on earth. Ambient air pollution is now recognized as one of the leading global threats to health and welfare. Fine particulate matter (PM_{2.5}) alone is responsible for millions of premature deaths each year, with especially high disease burdens in low- and middle-income countries (LMICs) in Asia and Africa (Health Effects Institute and Institute for Health Metrics and Evaluation, 2025).¹ It is also increasingly understood to have serious impacts on cognitive performance (Deryugina et al., 2019; Bishop et al., 2022; Pope III et al., 2020; La Nauze and Severnini, 2021; Bedi et al., 2021; Archsmith et al., 2018; Huang et al., 2020; Dong et al., 2021; Qin et al., 2019). Those impacts imply that sustained exposure to dirty air may erode human capital, which is central to long-run growth and mobility (Gethin, 2025; Heckman and Mosso, 2014). It is this dimension of the impacts of pollution that we focus on here.

Specifically, we study primary-school students in rural Punjab, Pakistan, and examine how PM_{2.5} exposure during the school year affects their performance on standardized assessments. Our focus is on the rate of learning: how much skill a child acquires over the school year under different pollution conditions, rather than only how they perform on a single test taken on a dirty-air day. While prior work has shown that acute pollution shocks on exam days can depress performance (e.g. Ebenstein et al., 2016), we provide evidence on whether, and by how much, chronically elevated pollution during the school year slows human capital accumulation in an LMIC setting characterized by very high ambient exposure, limited mitigation technologies, and constrained school resources.

To address this question, we bring together several large-scale data sources. Our education data come from the Learning and Educational Achievements in Punjab Schools (LEAPS) project,

¹Exposure is not only widespread but also unequally distributed: poorer countries tend to face much higher concentrations than their richer counterparts (Rentschler and Leonova, 2023). Pakistan is emblematic of this problem. It regularly ranks among the most polluted countries in the world in terms of population-weighted annual PM_{2.5}, with typical concentrations many times higher than the World Health Organization guideline value (Health Effects Institute and Institute for Health Metrics and Evaluation, 2025; Mir et al., 2022). Pakistan is an integral part of the Indo-Gangetic Plains and Himalayan Foothills (IGP-HF) airshed, a contiguous atmospheric basin spanning Bangladesh, India, Nepal, and Pakistan (The World Bank, 2022). This airshed is highly transboundary, with emissions and secondary pollutants transported across national borders through shared meteorological and topographical conditions. Across the IGP-HF region, crop residue burning is identified as a major seasonal source of particulate matter and precursor emissions, contributing significantly to air-quality deterioration in Pakistan as well as in neighboring countries (Heger et al., 2025).

a longitudinal study that tracks cohorts of students in public and private primary schools across more than 100 villages in Punjab (Andrabi et al., 2007). LEAPS administers repeated standardized tests in Urdu, English, and mathematics that are closely aligned with the national curriculum, and records rich information on students, schools, and households, exact school locations, and exam dates. We link these data to monthly gridded $PM_{2.5}$ estimates at fine spatial resolution derived from satellite retrievals and chemical transport models (van Donkelaar et al., 2021), to satellite detections of open fires, and to reanalysis-based measures of local meteorology, including wind direction and speed.

Our empirical strategy exploits the interaction of fires and wind to generate plausibly exogenous variation in school-year air quality. On any given day, pollution from some fires in the areas surrounding schools is carried toward a school by prevailing winds, while smoke from others is carried away. By combining information on fire locations with high-frequency wind fields, we construct measures of “upwind fire exposure” for each school and use these as instruments for local $PM_{2.5}$ concentrations over the school year. Intuitively, the timing and location of fires, and the direction of wind on those days, shift local particulate pollution in ways that are unlikely to be systematically correlated with local economic conditions or other unobserved determinants of learning. Our main outcomes are measures of student achievement on the LEAPS assessments, including the share of more difficult questions answered correctly, which we interpret as capturing cumulative learning between grades.

We document three main findings. First, higher $PM_{2.5}$ exposure during the school year substantially depresses test performance. Our preferred instrumental variable estimates indicate that a $1\mu g/m^3$ increase in $PM_{2.5}$ lowers the share of difficult items answered correctly by 2.7 percentage points — roughly 13.5% of the 0.20 sample mean and approximately 0.16 standard deviations of the outcome (Table 2, column 3). The effects are larger in mathematics than in language subjects, consistent with the idea that more cognitively demanding tasks are especially sensitive to environmental stressors. Second, the timing of exposure matters: $PM_{2.5}$ during the school year is strongly predictive of poorer outcomes, whereas exposure during the summer break has little explanatory power. This pattern is consistent with a mechanism operating during instructional months; we interpret this as suggestive rather than definitive evidence that pollution impacts students via direct classroom impacts rather than through more general background health or environmental

pollution.² Our focus is the accumulation of knowledge over the school year rather than acute effects on performance on the day of an exam, though the framework accommodates both. We let a student's performance P be a function of four inputs — attendance (A), cognitive function (C), effort (E), and instructional quality (I):

$$P = P(A, C, E, I) \quad (1)$$

Each input can respond to the air quality Q that students and teachers experience.³ Substituting and differentiating yields the total derivative of performance with respect to air quality:

$$P = P(A(Q), C(Q), E(Q), I(Q)) \quad (2)$$

with the corresponding derivative:

$$\frac{dP(A, C, E, I)}{dQ} = \underbrace{\frac{\partial P}{\partial A} \left[\frac{\partial A}{\partial Q} \right]}_1 + \underbrace{\frac{\partial P}{\partial C} \left[\frac{\partial C}{\partial Q} + \frac{\partial C}{\partial E} \frac{dE}{dQ} \right]}_2 + \underbrace{\frac{\partial P}{\partial E} \left[\frac{\partial E}{\partial Q} \right]}_3 + \underbrace{\frac{\partial P}{\partial I} \left[\frac{\partial I}{\partial Q} + \frac{\partial I}{\partial E} \frac{dE}{dQ} \right]}_4 \quad (3)$$

The four bracketed terms trace four distinct channels from air quality to measured performance. Term 1 is the attendance channel: pollution keeps students out of school. Term 2 is the cognitive channel; its second piece, $\frac{\partial C}{\partial E} \frac{dE}{dQ}$, captures the possibility that students facing a direct cognitive penalty raise effort to offset it. Term 3 is an effort channel that operates independently of any cognitive effect. Term 4 runs through instructional quality: teachers exposed to pollution may be less effective, or they may lose instructional time to pollution-related disruption.⁴

The evidence for each channel is well established. Experimental and field studies show that pollution lowers effort and impairs performance on cognitively demanding tasks (La Nauze and Severnini, 2021; Bedi et al., 2021; Archsmith et al., 2018; Huang et al., 2020), with the sharpest effects on the most difficult problems (Ma et al., 2025); classroom air quality should therefore move

²Srivastava et al. (2026) study the effect of temperature on test performance in Ethiopia. The same structure applies to pollution, with air quality in place of temperature as the environmental stressor.

³We abstract from the distinction between ambient and classroom pollution; the two are highly correlated in Pakistan, and our exposure measures are ambient.

⁴This term also absorbs the attendance and cognitive effects experienced by teachers, as well as any instructional time lost to managing disruptive behavior; one could expand $\frac{\partial I}{\partial Q}$ to separate these sub-channels.

collected detailed information on all public schools and a census of private schools in these villages. Within each village, LEAPS tested students in grades 3, 4, and 5 in Urdu, English, and mathematics using standardized assessments closely aligned with the official curriculum. The surveys also gathered rich information on student demographics, household characteristics, parental education, and schooling choices, as well as school-level data on infrastructure, fees, teacher characteristics, and governance. School and village locations were geocoded, and the exact dates of testing were recorded, which allows us to link students and schools to time-varying measures of local environmental conditions.

A key feature of LEAPS is its panel structure. Baseline testing and surveys were conducted in 2003, with follow-up rounds over subsequent years that re-tested the same cohorts of students and updated school and household information. The rounds we analyze span 2003 through 2012. Because the LEAPS panel predates the most recent decade, the pollution levels, crop-burning practices, school infrastructure, and household energy use it captures may differ from current conditions; we therefore read our estimates as causal evidence on the pollution–learning relationship rather than as a description of present-day exposure. This design allows us to track the same children as they progress through primary school, observe transitions between schools and sectors (public vs. private), and measure changes in achievement over time. The resulting dataset contains repeated test scores for approximately 50,000 students nested within several hundred schools, providing a large student-level panel in which we can relate changes in achievement to variation in school-year exposure to air pollution, while conditioning on rich time-invariant and time-varying controls at the student, household, and school levels.

3.1.2 Air pollution data (PM_{2.5})

Our primary measure of ambient air pollution is fine particulate matter (PM_{2.5}), expressed in micrograms per cubic meter ($\mu\text{g}/\text{m}^3$). We use monthly gridded PM_{2.5} estimates from the Atmospheric Composition Analysis Group, which combine satellite-retrieved aerosol optical depth from multiple instruments with a chemical transport model to produce globally consistent surface PM_{2.5} concentrations at fine spatial resolution (van Donkelaar et al., 2021). The data are available globally from the late 1990s onward on a regular latitude–longitude grid; for our analysis we ex-

tract the value of $PM_{2.5}$ in a 5km buffer around each school in the LEAPS data. We then construct school-level exposure measures that correspond to the timing of educational activities: for each test administration, we compute the average $PM_{2.5}$ over the 12 months preceding the exam date, as well as separate averages for months when schools are in session and for vacation months. These measures summarize the cumulative pollution environment that students experience during the relevant learning period while preserving meaningful variation across schools, cohorts, and years.

There are substantial level differences in pollution across the schools in our sample. Figure 1a shows the annual average pollution level for three randomly selected schools from each of the districts in our sample over the years that our sample falls into. Pollution levels in the most polluted school are nearly double those in the least polluted school. The trends in pollution also differ. The schools in both Faisalabad and Rahim Yar Khan experience increasing pollution levels over our sample period. The school in Attock, on the other hand, has no (or a slight downward) trend in pollution levels. These patterns are generally consistent in our summer-specific measures of pollution.

This variation in pollution levels and trends across schools in our sample motivates our use of a within student empirical design. Our empirical approach does not rely on the level differences documented in Figures 1a and 1b. Rather, our fixed effects mean that we identify the impact of pollution by relying on deviations in experienced pollution within individual students. We proxy this at the school level in Figures 1c and 1d. Here we plot the annual deviations from the de-trended average level of pollution by school. These figures highlight that, despite substantial level differences in pollution, each school experiences annual deviations that are of relatively similar magnitudes. Further, these deviations do not appear to be broadly correlated. In some years one school experiences positive deviations while others experience negative deviations. It is these deviations that we rely on for identification, comparing test scores in a year with a more positive deviation to test scores with a more negative deviation, across the whole sample. Across our sample, annual $PM_{2.5}$ averages $54 \mu g/m^3$ (SD 13) and ranges from 31 to $83 \mu g/m^3$ across school-years; our within-student design identifies the effect from the year-to-year deviations around each school's own level, which are smaller than these cross-school differences. The estimated coefficient is therefore a marginal effect identified over this high-baseline range, and should not be mechani-

3.3 Empirical Approach

Local economic activity is a major driver of local air pollution and improvements (or declines) in economic activity that lead to increases (or decreases) in pollution may have consequences for educational achievement as well in ways that lead to spurious correlation between pollution and student performance. Results from an OLS approach would then be biased. To address this concern we estimate an instrumental variables (IV) approach where we instrument for total PM_{2.5} exposure of students by exploiting both variation in the occurrence of residue fires surrounding schools, as well as variation in wind direction that determines how many days in a given year a school is exposed to smoke from those residue fires.⁶ For each school-year we count the number of days in the exposure window with ≥ 1 upwind residue fire within R km and within a Θ -degree wind cone and instrument for average pollution levels at that school over the course of that year using this count of downwind fire days.

We define upwind fires as those that occur within a given buffer radius around the school and within a cone defined by the wind direction. Cones run parallel to the wind direction and have width that we define and vary in robustness checks. Our primary specification uses a buffer of 5km and a wind cone of 45 degrees. We provide a schematic for what our definition of upwind fires looks like in Figure 2. In robustness checks we also utilize information about fires we define as downwind and orthogonal – those that are located perpendicular to the school and prevailing wind direction – as control variables.

We implement this strategy with a 2SLS approach and estimate the following system of equations:

$$PM_{2.5its}^m = \phi \text{Upwind fires}_{it} + \eta Temp_{it} + \delta Prec_{it} + \kappa \mathbf{X}_{it} + \psi_s + \gamma_j + \nu_t + \epsilon_{it} \quad (4)$$

$$Y_{its} = \beta \widehat{PM_{2.5it}} + \nu Temp_{it} + \Delta Prec_{it} + \zeta \mathbf{X}_{it} + \Psi_i + \Gamma_j + \Upsilon_t + \epsilon_{it} \quad (5)$$

where the outcome variable Y_{it} is the LEAPS performance of student i in year t in subject s . We study outcomes aggregated across all subjects and individually in math, English, and Urdu.

⁶Our estimates capture the impact of variation in PM_{2.5} induced by changes in upwind fires, they are not a direct estimate of the total impact of upwind crop burning on student outcomes.

We use several different measures of performance, but our primary focus is on the percentage of difficult questions answered correctly across all subjects. We define a year as the 365 days prior to LEAPS take date for a given student. For example, the relevant year for the students who take the exam in February 2007 will include the 365 days prior to February 2007 through to February 2006, with the precise dates defined by the exam date. The variable $PM_{2.5it}$ denotes the average $PM_{2.5}$ level experienced by student i in year t . $\eta Temp_{it}$ and $\delta Prec_{it}$ are controls for the average maximum temperature and precipitation for student i in year t . $\mathbf{X}_{it}$ is a vector of controls that varies by specification and includes controls for other meteorological conditions, downwind and orthogonal fires, and student specific controls. We also include student ψ_i , school γ_j , and year v_t fixed effects. In all regressions we cluster errors at the school by class level.

In the second stage we instrument for pollution levels with the number of days with at least one upwind fire that occur in the previous year. The outcome of interest is the β on our predicted level of pollution ($\widehat{PM_{2.5it}}$).

The key identifying assumption in this approach is that the only way that smoke from residue fires impacts student performance is via the changes they induce in average $PM_{2.5}$ levels. The most obvious threat to this assumption is that places that experience more residue fires may also be experiencing better economic conditions. In this case the higher number of residue fires would be caused by more agricultural production, which generates an economic benefit for the students' families and consequently has implications for their academic performance. This concern is what motivates our use of wind direction in addition to the raw count of fires. While the number of fires might be related to economic conditions, the frequency with which a given fire is upwind of a school will depend on random variation in the weather, which we rely on to break any connection between the number of fires in the area and local economic conditions.

We show results of the first stage of these regressions when we present our overall results. In general our first stage results indicate that our instruments perform well. Across all sets of controls we find that fires are a strong predictor of average $PM_{2.5}$. The first-stage Kleibergen–Paap rk Wald F-statistic exceeds the conventional standard of 10 for strong instruments in most specifications, dropping just below it (to roughly 8.8) only in the most saturated specifications, which add downwind and orthogonal fire controls.

4 The Impact of Cumulative Pollution Exposure

We find strong evidence that exposure to pollution reduces student learning. A $1\mu g/m^3$ increase in $PM_{2.5}$ in the year leading up to taking a LEAPS test lowers the share of correct answers to difficult questions across all subjects by 2.7 percentage points — roughly 13.5% of the 0.20 sample mean and 0.16 standard deviations of the outcome (Figure 3 & Table 2, column 3). Once we control for weather conditions - which both influence pollution levels and have an independent impact on student performance (Park et al., 2020, 2021) - our results are stable across all combinations of controls that we examine. These include controls for test day conditions, the presence of temperature inversions (which can increase pollution levels), the number of downwind and orthogonal fires, and student specific controls.⁷

When we examine effects by subject we find the largest impacts in math, with effects that are roughly 40% larger than our impact across all subjects (a reduction of roughly 3.8 percentage points, Figure 4). The impacts in English and Urdu are, correspondingly, smaller than the impacts across all subjects. The effects we document for math remain statistically significant but those for English and Urdu are generally no longer statistically significant. Larger impacts on math is consistent with existing work that finds heat generates larger impacts in math (Srivastava et al., 2026) relative to the social sciences, short-term exposure to air pollution generates larger effects in math than in reading (Dang et al., 2025), and that air pollution generates larger impacts on cognitive tasks that rely more on problem solving (Bedi et al., 2021).

When we examine the impacts of pollution during the school year and summer separately, we find strongly suggestive evidence that the impacts we document are driven by pollution during the school year rather than the summer. In Figure 5 we show the estimates from our primary specification, where we do not distinguish between school year and summer pollution in the first bar. We then plot coefficient estimates from a version of our primary specification where we instrument for school year and summer pollution separately. While these estimates are both noisier than our estimate of the impact of annual pollution the point estimates strongly indicate that the impact of school year pollution is large, roughly 90% of our estimate for the impact of annual pollution levels, and negative. They are not significant at conventional levels but are significant

⁷We control for the enrollment status of the student, whether the student was tested at home, child's gender, and child's age.

increases the number of days that students report being absent by just over 7.7% (t-stat: 3.00, First stage F-stat: 22.33).¹⁰

To assess the extent to which attendance impacts contribute to our overall estimates we conduct two exercises. First, we re-estimate our primary specification but split the sample by whether students have experienced any absences in the previous 30 days. We find when we do so that the impact of a $1\mu g/m^3$ increase in annual $PM_{2.5}$ in the sample that experiences at least one absence is roughly 45% larger than in the sample that does not (-0.029 [t-stat:2.06] vs. -0.020 [t-stat: 0.74]). This exercise is motivated by the fact that we cannot observe absences over the full previous year but recognize that absences over the previous 30 days may be correlated with total absences over the full year. If that is true then experiencing an absence over the previous 30 days may serve as a good proxy for students whose health is more sensitive to pollution and so may have experienced more absences during the full year due to pollution exposure. Our results are consistent with that hypothesis. However, this is subject to the caveat that our variation in absences is non-random and, despite the fact that we show that absences are causally related to pollution exposure, this heterogeneity analysis cannot demonstrate causally that absences are leading to the increased impacts of pollution.

The second exercise is a back of the envelope calculation that measures how much of our total effect can potentially be explained by the absence channel. On average, students experience 1.8 absences over the previous 30 days. Assuming constant absences across the school year, that implies an average of between 15 and 18 absences over a full school year. An annual increase in $PM_{2.5}$ of $1\mu g/m^3$ then leads to roughly an additional day absent. In the LEAPS data, each additional day of absence in the previous thirty days is associated with a -0.0009 (t-stat: 2.03) percentage point reduction in the share of the most difficult questions answered correctly. Assuming that this relationship is stable with respect to full year absences, the additional absence induced by pollution exposure accounts for approximately 3% of the overall impact of pollution that we estimate.

We view this estimate as indicative of the order of magnitude of the absence channel, rather

¹⁰We use inversions because our measure of residue fires is not a good instrument at a monthly level. Inversions are a consequence of the differences in temperature between different atmospheric layers. Normally temperature declines monotonically as one moves from the surface of the earth up through the atmosphere. Inversions occur when the layers just above the surface layer are warmer than the surface layer. This results in less mixing between layers and less air movement at the surface. As a result inversions are strongly associated with higher pollution events and have frequently been used as instruments for pollution (Fu et al., 2021; Chen et al., 2022; Dechezleprêtre et al., 2019).

5km fires. The consistency in magnitudes across wind cones in the results also suggest that it is the distance of the fire, and not their angle from the school, that primarily determines the strength of the instrument.

4.3.2 Alternative outcome measures

We also test robustness to using alternative measures of student performance included in the LEAPS data. In addition to measuring the percent of difficult questions answered correctly the LEAPS data also records other Item Response Theory (IRT) based measures of student performance i.e. EAP and MLE. These measures utilize IRT to generate latent ability scores that are comparable across students, schools, and survey rounds. Rather than relying on raw scores, IRT scaling accounts for variation in item difficulty and test design, allowing achievement to be placed on a common metric over time and enabling analysis of learning trajectories, school and teacher value added, and learning gaps across school types. These IRT-based scores are typically standardized and are reported separately by subject, with differences interpreted in standard deviation units.

We find broadly similar results when we use these outcomes. Table 3 reports results by subject for each of the alternative outcomes. In all cases the point estimates indicate negative effects of pollution exposure. For both EAP and MLE outcomes the impacts on math are the largest and most significant. Impacts on English and Urdu remain statistically significant at 10% levels for EAP but are not statistically significant when using the MLE outcome. Math impacts remain significant at 5% with both outcome measures.

5 Conclusion

Using student-level data on test performance combined with an IV based on wind direction and residue fires, we demonstrate that exposure to higher levels of PM_{2.5} pollution in the 12 months prior to an exam reduces performance among Pakistani students. These effects are large: a $1\mu g/m^3$ increase in average exposure over the previous 12 months reduces performance on the most difficult test questions by 2.7 percentage points, roughly 13.5% of the mean and about 0.16 standard deviations, but they are consistent with results from other high-pollution locations. We also find

suggestive, though statistically imprecise, evidence that effects are larger for male and younger students — a pattern opposite to the most closely comparable study, which we treat as hypothesis-generating. Because our instrument is smoke from upwind crop fires, these estimates recover the effect of the fire-driven component of $PM_{2.5}$ variation; $PM_{2.5}$ in Punjab also reflects transport, industry, household energy, dust, secondary formation, and transboundary transport, of which crop-residue burning is one important and actionable seasonal source.

A few caveats bound these results. Our estimates are identified on primary-school students in grades 3 through 5, in rural communities across three districts of Punjab. They need not extrapolate to urban centers such as Lahore or Karachi, where pollution sources, school infrastructure, and socioeconomic conditions differ substantially; to secondary or older students; or to other provinces, and they should not be read as representative of Pakistan’s education system as a whole. The LEAPS panel also predates the most recent decade, so present-day exposure, burning practices, and school conditions may differ from those we study. Establishing whether comparable effects hold in urban and non-Punjab settings, and under current conditions, is a priority for future work.

Our results make several important contributions and open several avenues for future research. First, we add to the evidence that long-term exposure to elevated levels of pollution reduces student learning (Wen and Burke, 2022; Gilraine and Zheng, 2024; Martin et al., 2025). This effect is distinct from, and to the extent that it represents permanent reductions in human capital, more consequential than the impact that pollution has on test day performance (Marcotte, 2015; Ebenstein et al., 2016; Cavalcanti, 2024; Carneiro et al., 2021; Zivin et al., 2020; Andersen et al., 2025; Koivuranta et al., 2024). Our results are distinct from much of the existing work on pollution and learning because they come from a setting with extremely high average levels of pollution. Only one other study (Balakrishnan and Tsaneva, 2021) examining this question has measured the impact of pollution in a setting with even half the average level of pollution of our setting. We find similar total impacts, but different patterns of heterogeneity, in our context. Our results also add one more piece of evidence that long-term exposure to pollution does have impacts on student performance in LMICs, in contrast to some recent work suggesting pollution only has short-term effects (Dang et al., 2025).

Second, we provide some evidence for the channels through which pollution impacts student

performance. While we cannot fully isolate the channels through which pollution impacts performance, our evidence suggests that pollution-induced absences account for only a small portion of the overall impact—our back-of-the-envelope calculation puts the share well below 10%, though that figure rests on strong assumptions and should be read as indicative. That suggests, in line with existing work on the direct cognitive impacts of pollution (La Nauze and Severnini, 2021), that a substantial portion of the impact is due to non-attendance related learning losses, potentially, driven by direct impacts of pollution on cognition.

Third, our results highlight the potentially long-term consequences of air pollution for economic growth. The short-term, and long-term, health impacts of pollution are well documented. But our results indicate that pollution can lead to reductions in human capital accumulation, independent from their impacts on health, that may have long-term growth consequences. One important area of future research will be converting the impact of pollution on learning losses into impacts on long-term growth rates. Doing so will help provide a clearer estimation of the potential economic benefits of pollution regulation.

In addition to determining the impact of pollution driven learning losses on long-term growth rates, our study raises several questions for future research. Our findings differ in their patterns of heterogeneity from those reported by Balakrishnan and Tsaneva (2021), who document larger impacts among female and older students, whereas we find suggestive evidence of larger effects for male and younger students. These differences may reflect variation in baseline exposure levels, institutional contexts, or identification strategies, and highlight the need for further work to understand how the distributional impacts of pollution on learning vary across settings. A second question is whether the effects of pollution persist in higher-stakes exams. None of the exams we, or Balakrishnan and Tsaneva (2021), examine are high stakes. The literature on the learning impacts of heat exposure has demonstrated that the impact of heat on high and low-stakes exams can differ markedly. It remains to be determined if that is true in the case of pollution as well.

5.1 Policy implications

Our findings carry several implications for protecting learning during high-pollution periods. Because much of the estimated effect appears to operate during instructional months and through

channels other than absence, policies that track attendance alone will not be sufficient. Schools and provincial authorities could establish protocols for severe-pollution days and, where feasible, deploying low-cost classroom air filtration. Integrating air-quality information into school health and attendance systems would help education authorities identify the schools and weeks of greatest exposure and target scarce resources accordingly. Realizing these gains would require coordination across the environment, education, health, and agriculture departments, which at present manage air quality, schooling, and the agricultural calendar largely in isolation. One important unresolved question, especially in a context where low-cost purifiers are deployed in classrooms, is whether policies that cancel classes on high pollution days cause more harm than they prevent. Canceling classes on high pollution days reduces the effective learning days that students have and may reduce human capital formation. If exposure to pollution is higher outside of the classroom, these losses may come with no corresponding benefit to students from reduced exposure. If classrooms are equipped with low-cost purifiers, it may be the case that canceling class both reduces learning and increases health impacts by substantially increasing exposure.

Beyond schools, a natural response is to curb the upwind crop-residue burning that drives the seasonal exposure we exploit. However, crop-residue burning is only one of many sources of ambient PM_{2.5} in Punjab (Heger et al., 2025), so even eliminating it would address only part of total exposure. Regulating burning is also difficult: farmers face tight windows between harvest and the next sowing, the alternatives to burning are costly relative to smallholder margins, and earlier regulatory attempts in the region have achieved limited compliance. Durable reductions would likely require addressing these underlying incentives through subsidized or shared residue-management equipment, viable uses for crop residue, and alignment of the agricultural calendar with air-quality goals.



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Nutrition, information and household behavior: Experimental evidence from Malawi



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ABSTRACT

Incorrect knowledge of the health production function may lead to inefficient household choices and thereby to the production of suboptimal levels of health. This paper studies the effects of a randomized intervention in rural Malawi that, over a six-month period, provided mothers of young infants with information on child nutrition without supplying any monetary or in-kind resources. A simple model first investigates theoretically how nutrition and other household choices including labor supply may change in response to the improved nutrition knowledge observed in the intervention areas. We then show empirically that the intervention improved child nutrition, household food consumption and consequently health. We find evidence that labor supply increased, which might have contributed to partially fund the increase in food consumption. This paper is the first to study whether non-health choices, particularly parental labor supply, might be affected by parents' knowledge of the child health production function.

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1. Introduction

Since Becker's (1965) seminal contribution, economists have recognized that many goods are not directly bought in the market, but are produced at home using a combination of market and non-market goods. The home production framework has been particularly fruitful in studying the production of health, in particular child health (Gronau, 1986, 1997; Grossman, 1972; Rosenzweig and Schultz, 1983). An important implication of such models is that households make choices given their knowledge of the (child) health production function. Consequently, deficiencies in knowledge lead to suboptimal household choices and thereby distorted levels of child health. Establishing empirically the consequences of deficiencies in knowledge on household behavior has, however, been challenging because knowledge is endogenous and is usually either unobserved or proxied by education, which also affects child health through other channels including earnings.

In this paper, we overcome this challenge by exploiting an intervention, implemented through a cluster randomized trial, aiming to improve mothers' knowledge of the child health production function in rural Malawi. The intervention solely provided information on child nutrition to mothers, thus yielding a clean source of identification.

Our contribution is twofold. First, we assess whether the intervention improved child nutrition and consequently health. Second, drawing on a simple theoretical model, we investigate how other household choices change to accommodate the improved knowledge of the production function. In so doing, we assess whether non-health choices, particularly parental labor supply, might be affected by parents' knowledge of the child health production function.

In the context we study, rural Malawi, mothers have many misconceptions about child nutrition. To take some examples, it is common practice to give porridge diluted with unsterilized water to infants as young as one week; the high nutritional value of groundnuts, widely available in the area, is not well known; and widespread misplaced beliefs include that eggs are harmful for infants as old as 9 months and that the broth of a soup contains more nutrients than the meat or vegetables therein. This evidence suggests that important changes can be expected if these misconceptions are corrected.

The intervention we study delivered information in an intense manner: trained local women visited mothers in their homes once before the birth of their child and four times afterwards and provided information on early child nutrition on a one-to-one basis. Moreover, the fact that the intervention had been running for at least three years when outcome data were collected allows a sufficient time frame for practices to change. This lapse also allows us to measure medium-term impacts, which is important since interventions often perform much better in the short rather than medium term (Banerjee et al., 2008; Hanna et al., 2016).

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Consistent with gains in knowledge, we find evidence of improvements in infants' diets and household food consumption, particularly an increase of protein-rich foods and of fruit and vegetables. We also find that household food consumption increases and there is suggestive evidence that this might have been partially financed through increased labor supply. Overall, the findings are consistent with households learning that some relatively costly foods are more nutritious than they previously believed and adjusting their labor supply so as to facilitate increases in their children's intake of them. Indeed, we show that households adjust their behavior on several margins including child diet inputs and labor supply, making their response more complex than simply changing the composition of consumption while keeping total consumption constant.

We find that the intervention improved children's physical growth, particularly height, a widely used indicator of long-term nutritional status. This finding is particularly important for policy: child malnutrition is a severe and prevalent problem in developing countries (de Onis et al., 2000) and leads to poor health and excess child mortality (Bhutta et al., 2008; Pelletier et al., 1994) and is also linked to poor human capital outcomes later on in life.¹

The paper deals carefully with the increasingly important issue of inference in cluster randomized trials when the number of clusters is small. It is well known that in this situation, standard statistical formulae for clustered standard errors based on asymptotic theory (cluster-correlated Huber–White estimator) provide downward-biased standard error estimates (Bertrand et al., 2004; Cameron et al., 2008; Donald and Lang, 2007; Wooldridge, 2004). We use two leading methods for inference in this case – randomization inference (Fisher, 1935; Rosenbaum, 2002) and wild-cluster bootstrap-t (Cameron et al., 2008). Furthermore, we assess their performance in our data using Monte Carlo experiments and find that both methods perform relatively well. Presenting the performance of these two methods side-by-side is of interest for many empirical applications, given the increasing trend in randomized trials with a small number of clusters.

Lewycka et al. (2013) study the effect of the intervention we study on exclusive breastfeeding and infant mortality. Our paper addresses a different question: whether improving knowledge of the health production function affected consumption, labor supply, nutritional practices and child nutrition to the age of around 5 years. We also use a different dataset; they interview mothers until their child is 6 months old, while we rely on a representative sample of women of reproductive age and their households. More details about the design of the intervention can be found in Lewycka et al. (2010).

Our work contributes to a number of strands of literature. First, it adds to the discussion on the effects of health information on behavior (Dupas, 2011a).² The evidence is mixed: Dupas (2011b); Jalan and Somanathan (2008) and Madajewicz et al. (2007) find that providing information on, respectively, the risks of contracting HIV and the arsenic and fecal concentration of water improves associated practices, while Kamali et al. (2003); Kremer and Miguel (2007) and Luo et al. (2012) find that health behaviors relating to, respectively, HIV, deworming and anemia do not respond to health education. This paper departs from these studies by not only considering a multifaceted information intervention, but also by studying household responses on a wider range of margins than those directly targeted by the intervention. In doing so, this is one of the first papers to investigate how behaviors not directly related to the topic of an information campaign adjust to it.

Second, this paper contributes to the literature evaluating the effects of nutrition information interventions on nutrition practices and child health. Haider et al. (2000) and Morrow et al. (1999) find increased rates of exclusive breastfeeding within small-scale randomized control trials in Bangladesh and Mexico respectively, while Alderman (2007); Galasso and Umapathi (2009) and Linnemayr and Alderman (2011) find improvements in child weight-for-age, an indicator for medium-term health status, using non-experimental methods. Our paper builds on these by studying the effects on a range of measures of child health, health practices and other margins of household behavior, all identified through a randomized control trial.

Finally, it relates to the literature investigating the causal effects of parental education on child health. In developed countries, Currie and Moretti (2003) and McCrary and Royer (2011) find, respectively, decreased incidence of low birth weight and modest effects on child health of increased maternal schooling in the US, while Lindeboom et al. (2009) find little evidence that parental schooling improves child health in the UK. For developing countries, Breierova and Duflo (2004) and Chou et al. (2010) find that parental schooling decreases infant mortality in Indonesia and Taiwan respectively. However, it is difficult to disentangle whether the effect of education is working through changes in knowledge of the child health production function, or through increased income and hence access to more and better-quality care. Related to this, Glewwe (1999) and Thomas et al. (1991) find that almost all of the impact of maternal education on child's height in Morocco and Brazil can be explained by indicators of access to information and health knowledge.

The rest of the paper is structured as follows. Section 2 provides background information on rural Malawi and describes the experimental design and data, Section 3 describes the theoretical framework and Section 4 sets out the empirical model. Our main results are presented in Section 5. Section 6 rules out alternative potential explanations behind our findings, while Section 7 concludes.

2. Background and intervention

2.1. Background

Malnutrition in the early years (0–5) is one of the major public health and development challenges facing Malawi, one of the poorest countries in sub-Saharan Africa. The 2004 Malawi Demographic and Health Survey (DHS) Report indicates an under-5 mortality rate of 133 per 1000, and under-nutrition is an important factor driving this: Pelletier et al. (1994) estimate that 34% of all deaths before age 5 in Malawi are related to malnutrition (moderate or severe). Moreover, 48% of Malawian children aged under 5 suffer from chronic malnutrition, a rate that is the second highest in sub-Saharan Africa.

Poor feeding practices are at least partly responsible for these extreme malnutrition indicators. Over half of all infants aged under 6 months are given food and/or unsterilized water (2004 DHS Report), which can lead to gastrointestinal infections and growth faltering (Haider et al., 2000; Kalanda et al., 2006) and is contrary to the World Health Organization (WHO) recommendation of exclusive breastfeeding for the first 6 months of an infant's life. Furthermore, porridge diluted with unsterilized water is often given in large quantities to infants as young as 1 week (Bezner-Kerr et al., 2007). In terms of nutrition for infants aged over 6 months, their diets – rich in staples such as maize flour – frequently lack the necessary diversity of foods to provide sufficient amounts of energy, proteins, iron, calcium, zinc, vitamins and folate: in our sample, 25% of children aged 6–60 months did not consume any proteins over the three days prior to the survey, with a further 30% consuming just one source of protein. Poor nutritional practices are likely to be related to a lack of knowledge: for instance, only 15% of mothers in our sample knew how to best cook fish combined with the local staple so as to maximize nutritional value.

It is against this background that, in 2002, a research and development project called MaiMwana (Chichewa for 'Mother and Child') was

¹ See, among others, Alderman et al. (2001); Almond and Currie (2011); Banerjee et al. (2010); Barham (2012); Behrman (1996); Behrman and Rosenzweig (2004); van den Berg et al. (2006, 2009, 2010); Bhalotra et al. (2015); Currie (2009); Currie et al. (2010); Glewwe et al. (2001); Hoddinott et al. (2008); Lindeboom et al. (2010); Maccini and Yang (2009); Maluccio et al. (2009); Schultz (2005) and Strauss and Thomas (1998).

² For the case of education, see for instance Jensen (2010).

set up in Mchinji District, in the Central region of Malawi.³ Its aim was to design, implement and evaluate effective, sustainable and scalable interventions to improve the health of mothers and infants. Mchinji is a primarily rural district, with subsistence agriculture being the main economic activity. The most commonly cultivated crops are maize, groundnuts and tobacco. The dominant ethnic group in the district is the Chewa (over 90% in our data). According to the 2008 Malawi census, socio-economic conditions are comparable to or poorer than the average for Malawi (in parentheses in what follows), with literacy rates of just over 60% (64%), piped water access for 10% (20%) of households and electricity access for just 2% (7%) of households.

2.2. The intervention

In 2005, MaiMwana established an infant feeding counseling intervention in Mchinji District (ongoing at time of follow-up), to impart information and advice on infant feeding to mothers of babies aged under 6 months.⁴ The intervention thus targets the very first years of life, a critical period for growth and development, during which nutritional interventions are likely to be most beneficial (Schroeder et al., 1995; Shrimpton et al., 2001; Victora et al., 2010). The information is provided by trained female volunteers ('peer counselors' hereafter) nominated by local leaders. In practice, peer counselors are literate local women aged 23–50 with breastfeeding experience.⁵

Each peer counselor covers an average population of 1000 individuals, identifying all pregnant women within this population and visiting them five times in their homes: once before giving birth (third trimester of pregnancy) and four times afterwards (baby's age 1 week, 1 month, 3 months, 5 months). Although all pregnant women are eligible for the intervention and participation is free, in practice around 60% of them are visited by the peer counselors. Our data show that women who were visited by the peer counselor tend to be poorer: in particular, they were 4.8 percentage points (7.5 percentage points) less likely to have a floor (roof) built with good materials.

Exclusive breastfeeding is strongly encouraged in all visits. Information on weaning is provided from when the baby is 1 month old (visits 3–5) and includes suggestions of suitable locally available nutritious foods, the importance of a varied diet (particularly the inclusion of protein and micronutrient-rich foods, including eggs) and instructions on how to prepare foods so as to conserve nutrients and ease digestion (e.g. to mash vegetables rather than liquidize them and to pound fish before cooking it). Peer counselors were provided with a manual to remind them of the content relevant for each visit and with simple picture books to aid in explaining concepts.

2.2.1. Experimental design

The evaluation is based on a cluster randomized control trial designed as follows (see Lewycka (2011) and Lewycka et al. (2010, 2013)). Mchinji District was divided into 48 clusters by combining enumeration areas of the 1998 Malawi Population and Housing Census (National Statistical Office, Malawi, 2008).⁶ This was done in a systematic way, based on the contiguity of enumeration areas and respecting

boundaries of Village Development Committees (VDCs), such that each cluster contained approximately 8000 individuals. Within each cluster, the 3000 individuals (equating to 14 villages on average) living closest to the geographical center of the cluster were chosen to be included in the study.⁷ The study population therefore comprises of individuals living closest to the geographical center of the clusters and was selected in this way in order to limit contamination between neighboring clusters by creating a natural buffer area. Twelve clusters were randomly selected to receive the infant feeding counseling intervention, with an average of three peer counselors per cluster. A further 12 serve as controls.⁸

2.2.2. Evaluation sample description

A census of women of reproductive age was conducted by MaiMwana in all clusters in 2004 ('baseline census' hereafter), before the intervention started in July 2005 (see Fig. 1).⁹ Approximately 3.5 years into the intervention, which was still in place, we drew a random sample from the baseline census in order to conduct the first follow-up survey.¹⁰ Specifically, in 2008 we drew a random sample of 104 women of reproductive age (17–43), regardless of their childbearing status,¹¹ from each of the 24 clusters, leaving us with a target sample of 2496 women. The baseline census contains some socio-economic and demographic characteristics of these women and their households, as shown in the left panel of Table 1. Women are on average 24.5 years old, just over 61% of them are married, over 70% have some primary schooling but just 7% have some secondary schooling. Households are predominantly agricultural and poverty is high, as indicated by the housing materials and assets. The table also shows that the randomization worked well, with the sample well balanced across intervention and control clusters at baseline given that only 1 out of 24 variables turns out to be unbalanced.¹²

We assess the impact of the intervention over 3.5 years after it began. While this has the benefit of allowing us to assess the effect of the intervention in the medium rather than short term, it also increases the risk of attrition. We succeeded in interviewing around two-thirds of the sample drawn for the first follow-up survey: 65% in intervention clusters and 68% in control clusters. Apart from the time lapse between baseline and the first follow-up, two additional factors contributed to the attrition. First, the district of Mchinji is particularly challenging for the collection of panel data because respondents are known to report 'ghost members' – fictitious household members – with the intention of increasing future official aid/transfers that may depend positively

⁷ The geographic center was chosen to be the most central village in the cluster as shown on a cartographic map from the National Statistical Office, Malawi. See Lewycka (2011, p. 122) for more details.

⁸ Another 24 clusters were randomly assigned to receive a participatory women's group intervention, whereby women of reproductive age were encouraged to form groups to meet regularly to resolve issues relating to pregnancy, childbirth and neonatal health. Child nutrition was not a primary focus and so we exclude these clusters from this analysis (see, instead, Rosato et al. (2006, 2009) and Lewycka et al. (2013)). MaiMwana Project also improved health facilities across the District, which equally benefitted intervention and control clusters.

⁹ Further details on this baseline census can be found in Lewycka et al. (2010). We take the intervention start date to be July 2005, the date by which the first six-month cycle had been fully completed, in line with Lewycka et al. (2013).

¹⁰ Data collection was carried out by MaiMwana in collaboration with the authors. Data were collected in Nov-2008–Mar-2009 (Oct-2009–Jan-2010) at first (second) follow-up using PDAs. To ensure that results were not driven by seasonality, field teams collected data in intervention and control clusters at the same time. The data are available for download at <https://discover.ukdataservice.ac.uk/catalogue/?sn=6996&type=Data%20catalogue>.

¹¹ This was done to avoid any potential bias arising from endogenous fertility decisions in response to the intervention. This turns out not to be an important concern, as we show in Section 6.

¹² Other welfare programs were operating in the District at the same time as this intervention. The potentially most important is the Mchinji Social Cash Transfer, providing cash transfers to the poorest 10% of households in the District. At follow-up, the intervention was in the pilot stage and only 2.5% of households in our sample (distributed evenly between intervention and control clusters) report having received it.

³ MaiMwana is a Malawian trust established as a collaboration between the Department of Pediatrics, Kamuzu Central Hospital, the Mchinji District Hospital and the UCL Centre for International Health and Development. See <http://www.maimwana.malawi.net/MaiMwana/Home.html>.

⁴ Though the intervention is predominantly focused on nutrition, it also touches on other issues such as birth preparedness, HIV testing and counseling, vaccinations and family planning. Section 6 discusses how these aspects relate to our results.

⁵ Peer counselors receive an initial five-day and annual refresher training, and attend monthly meetings. They are not paid, but receive a bicycle, meeting allowances, registers, calendars and supervision forms. They are supervised by 24 government health surveillance assistants and three MaiMwana officers.

⁶ The District Administrative Center was excluded because it is relatively more urbanized and less comparable to the rest of the District.

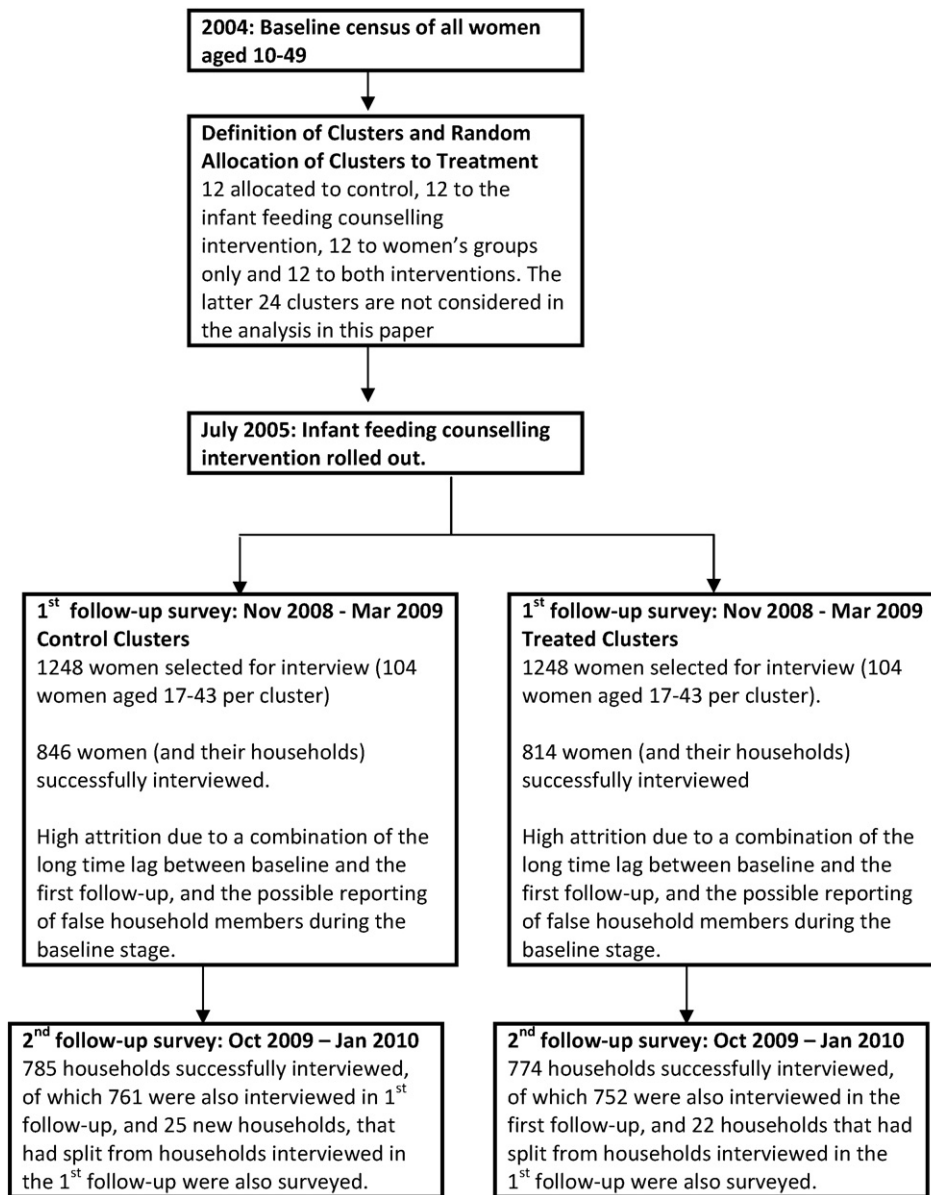


Fig. 1. Flowchart.

on household size (see Miller and Tsoka (2012) for more on 'ghost members' and Giné et al. (2012) for problems relating to personal identification in Malawi). Hence, it is possible that some women listed in the baseline census were in fact 'ghost members' and so could not be found by the field team in 2008. Second, an unexpected sharp drop of the British Pound against the Malawi Kwacha resulted in fewer resources to track women who had moved.

The middle panel of Table 1 shows that the balance on baseline characteristics is maintained in the sample of women who were found ('interviewed sample'). A small imbalance is detected on just 1 variable at the 10% level, suggesting that attrition between baseline and first follow-up was not significantly different between intervention and control clusters. While this is reassuring, it could nonetheless be the case that there is differential attrition in terms of unobserved variables. We dispel these concerns in Appendix A. We conducted a second follow-up survey of these women one year later, in 2009–10, successfully interviewing around 94% of the women interviewed at first follow-up: 95% and 93% in intervention and control areas respectively. The balance on baseline

characteristics for this sample, displayed in the right panel of Table 1, is very similar to that for the first follow-up.

The surveys contain detailed information on household consumption; consumption of liquids and solids for each child in the household (≤ 6 years); breastfeeding practices (≤ 2 years); health for all individuals in the household, reported by main respondent; weights and heights of children (≤ 6 years); labor supply (≥ 6 years); and the main respondent's knowledge about child nutrition.

3. Conceptual framework

In order to understand how information of the type provided by the intervention might affect household decisions, we present a simple theoretical model in which households care about adult consumption and leisure and about their child's health, which is a function of the child's consumption of a combination of nutrition inputs. For simplicity, we assume that this is a bundle of two inputs, C_1 and C_2 . We also assume that households have one adult and one child. The adult chooses simultaneously the amounts to spend on each child consumption

Table 1
Baseline sample balance.

	Full sample			Interviewed sample – wave 1			Interviewed sample – wave 2		
	Control group	Difference: treatment–control	p-Value	Control group	Difference: treatment–control	p-Value	Control group	Difference: treatment–control	p-Value
<i>Woman's characteristics</i>									
Married (dummy variable = 1)	0.615	−0.021	0.386	0.661	−0.034	0.184	0.654	−0.024	0.340
Some primary schooling or higher	0.707	0.033	0.402	0.682	0.040	0.340	0.68	0.037	0.438
Some secondary schooling or higher	0.066	0.010	0.535	0.060	−0.007	0.545	0.059	−0.006	0.607
Age (years)	24.571	−0.180	0.637	25.492	−0.429	0.376	25.397	−0.217	0.621
Chewa	0.948	−0.044	0.330	0.957	−0.050	0.246	0.959	−0.054	0.268
Christian	0.977	0.006	0.476	0.979	0.008	0.336	0.981	0.005	0.454
Farmer	0.661	−0.075	0.108	0.688	−0.060	0.128	0.678	−0.055	0.220
Student	0.236	0.015	0.438	0.204	0.022	0.274	0.208	0.017	0.410
Small business/rural artisan	0.036	0.030	0.129	0.037	0.024	0.220	0.039	0.025	0.264
<i>Household characteristics</i>									
Agricultural household	0.995	−0.005	0.471	0.995	0.002	0.591	0.995	0.003	0.500
Main flooring material: dirt, sand or dung	0.913	−0.041	0.232	0.916	−0.027	0.474	0.916	−0.028	0.422
Main roofing material: natural material	0.853	−0.018	0.697	0.857	−0.004	0.891	0.86	−0.008	0.861
HH members work on own agricultural land	0.942	−0.057	0.124	0.950	−0.056	0.120	0.95	−0.06	0.140
Piped water	0.011	0.040	0.314	0.009	0.032	0.340	0.01	0.034	0.440
Traditional pit toilet (dummy variable = 1)	0.772	0.054	0.218	0.791	0.054	0.182	0.796	0.044	0.324
# of HH members	5.771	0.066	0.817	5.848	0.132	0.863	5.903	0.096	0.833
# of sleeping rooms	2.116	0.199	0.038*	2.152	0.166	0.128	2.174	0.155	0.136
HH has electricity	0.002	0.007	0.166	0.002	0.004	0.338	0.003	0.004	0.394
HH has radio	0.630	0.030	0.408	0.641	0.015	0.709	0.645	0.014	0.655
HH has bicycle	0.509	0.015	0.643	0.512	0.008	0.843	0.512	0.01	0.769
HH has motorcycle	0.008	0.001	0.925	0.007	0.002	0.779	0.008	0.003	0.685
HH has car	0.006	−0.002	0.612	0.007	−0.003	0.298	0.008	−0.004	0.302
HH has paraffin lamp	0.925	0.032	0.262	0.926	0.036	0.178	0.935	0.026	0.360
HH has oxcart	0.058	−0.015	0.204	0.059	−0.022	0.090+	0.06	−0.022	0.072+
N	1248	1248		846	814		785	774	

Notes to table: p-values are computed using the wild-cluster bootstrap-t procedure as in [Cameron et al. \(2008\)](#), explained in [Section 4.1](#). 'Full sample' includes all women (and their households) originally drawn to be part of the 2008–09 survey. 'Analysis sample – wave 1' includes women (and their households) who were interviewed in 2008–09 (wave 1), while 'Analysis sample – wave 2' includes women (and their households) who were interviewed in 2009–10 (wave 2).

* p < 0.05.

** p < 0.01.

+ p < 0.1.

input, C_1 and C_2 , adult consumption, A , and leisure, L (or labor supply, $T - L$, where T is total time endowment of the adult). The household's optimization problem is therefore:

$$\begin{aligned} \max_{\{C_1, C_2, A, L\}} & U(A, L, H) \\ \text{st: } & A + p_1 C_1 + p_2 C_2 \leq w(T - L) \\ & H = F(C_1, C_2) \end{aligned}$$

where $U(\dots)$ captures the utility from adult consumption, leisure and child health, p_1 and p_2 are the prices of child nutrition inputs relative to adult consumption, and w is the wage per unit of time.¹³ The function $F(\dots)$ represents the health production function, which is increasing in both C_1 and C_2 , and concave. Following [Cunha et al. \(2013\)](#) and [Del Boca et al. \(2014\)](#), we assume that both the utility function and the production function are Cobb–Douglas, i.e. $U(A, L, H) = A^\alpha L^\beta H^\gamma$ and $H = C_1^\delta C_2^\theta$, with $\alpha, \beta, \gamma, \delta, \theta > 0$ and $\delta + \theta < 1$. We can therefore rewrite the optimization problem as:

$$\begin{aligned} \max_{\{C_1, C_2, A, L\}} & A^\alpha L^\beta C_1^{\gamma_1} C_2^{\gamma_2} \\ \text{st: } & A + p_1 C_1 + p_2 C_2 \leq w(T - L) \end{aligned}$$

where $\gamma_1 = \gamma\delta$ and $\gamma_2 = \gamma\theta$.¹⁴

¹³ We use a static, unitary model to draw out the key behavioral responses to the intervention in the simplest way. See [Blundell et al. \(2005\)](#) and [Chiappori \(1997\)](#), among others, for work that incorporates labor supply, household production and/or children within a collective framework. See [Grossman \(1972\)](#) for dynamic considerations of a health production function.

¹⁴ We assume that the household cannot borrow, which is consistent with well-known credit constraints in developing countries, as discussed for instance in [Dupas \(2011a\)](#).

Households make their consumption and labor decisions under their own perception of the child health production function, $C_1^\delta C_2^\theta$, which might differ from the true one (see [Cunha et al. \(2013\)](#)). This perceived production function depends on δ and θ , two parameters that measure the household's perception of the returns to child nutrition inputs. Changes in these parameters will change γ_1 and γ_2 .

To study the effect of the intervention, we differentiate the first-order conditions with respect to γ_1 (see [Appendix B](#)) and find that $\frac{dC_1}{d\gamma_1} > 0$, but that $\frac{dC_2}{d\gamma_1} < 0$, $\frac{dA}{d\gamma_1} < 0$ and $\frac{dL}{d\gamma_1} < 0$. This allows us to establish the following proposition:

Proposition 1. If γ_1 increases, then C_1 and total household consumption increase, but C_2 , A and L decrease. Similarly, if γ_2 increases, then C_2 and total household consumption increase, but C_1 , A and L decrease.

The intuition is as follows. If the perceived productivity of C_1, γ_1 , increases, then more will be consumed of this input. Given the concavity of the utility function, this increase is better accommodated by a small decrease in all other arguments of the utility function (C_2, A and L) than by a large decrease in only one of them. Note that the increase in C_1 is not fully offset by the decrease in C_2 and A , because L also decreases, which implies that labor supply increases. As there is no borrowing or saving, the increase in labor supply implies an increase in overall household consumption.¹⁵

The intervention promotes the consumption of protein-rich foods, fruit and vegetables relative to others such as staples. If C_1 summarizes

¹⁵ Our simple model abstracts from differential labor supply responses of mother and father. In a two-parent model, one could imagine that additional time devoted to the acquisition and preparation of more nutritious foods might be to the detriment of the mother's labor supply and/or leisure.

the goods that the intervention promotes and C_2 summarizes the consumption of staples, then the effect of the intervention can be summarized in terms of increasing γ_1 but decreasing γ_2 . Following Proposition 1, we expect an important composition effect (an increase in C_1 and a decrease in C_2) but the predictions on labor supply and adult and total consumption are in principle ambiguous because these will depend on whether the γ_1 or the γ_2 effect dominates. This is ultimately an empirical issue, which we study below.

4. Empirical framework

4.1. Estimation and inference

The randomized experiment provides a clean and credible source of identification to test the proposition emerging from the theoretical framework above. To do so, we estimate OLS regressions of the form

$$Y_{ict} = \alpha + \beta_1 T_c + X_{ict}\beta_2 + Z_{c0}\beta_3 + \mu_t + u_{ict}, \quad t = 1, 2 \quad (1)$$

where Y_{ict} includes outcomes for unit i (household or individual, depending on the outcome of interest) living in cluster c at time t ($= 1, 2$ for first and second follow-ups, 2008–09 and 2009–10, respectively).¹⁶ In line with the model, the dimensions of household behavior likely to be affected include household and child consumption, labor supply and child health;¹⁷ T_c is a dummy variable that equals 1 if the main respondent of our survey was, at the time of the baseline in 2004, living in a cluster that later received the intervention; X_{ict} is a vector of household/individual-level variables measured at time t including a quadratic polynomial in age and gender; Z_{c0} is a vector of cluster-level variables measured at baseline such as proportions of women with Chewa ethnicity and proportions with primary or secondary schooling; μ_t is a vector of month-year dummies indicating the month of the interview; and u_{ict} is an error term, which is uncorrelated with the error term of others living in other clusters ($E[u_{ict}u_{jwq}] = 0$ for $i \neq j, c \neq w$) but which may be correlated in an unrestricted way with that of others living in the same cluster, independently of the time period ($E[u_{ict}u_{jct}] \neq 0$). Note that this correlation structure allows for the error term for individuals/households in the same cluster to be correlated over time, and also for the presence of spillovers within but not across clusters, which is reasonable for our case given the presence of large buffer areas in place between study areas in adjacent clusters, as discussed in Section 2.2.1.

The treatment indicator, T_c , takes the value 1 if the respondent was living in a treatment cluster at the time of the 2004 baseline census and 0 if living in a control cluster at that time. Therefore, we identify an intention-to-treat parameter. Moreover, defining T_c on the basis of baseline rather than current residence circumvents any bias that might arise from selective migration from control to treatment clusters.

In terms of inference, standard statistical formulae for clustered standard errors based on asymptotic theory (cluster-correlated Huber–White estimator) provide downward-biased standard error estimates if the number of clusters is small, thus over-rejecting the null of no effect (Bertrand et al., 2004; Cameron et al., 2008; Donald and Lang, 2007; Wooldridge, 2004). This is a potential issue, as there are just 24 clusters. We use two approaches proposed to obtain valid inference: wild cluster bootstrap-t (Cameron et al., 2008) and randomization inference (Fisher, 1935; Rosenbaum, 2002).

To implement randomization inference, we follow Small et al. (2008) to account for covariates by regressing the outcome variable on all covariates, except for T_c , and applying the randomization

inference procedure to the residuals from this regression. The test statistic is as follows:

$$\sum_{c:T_c=1} \frac{\hat{v}_{ict}}{N_1} - \sum_{c:T_c=0} \frac{\hat{v}_{ict}}{N_0}$$

where $\hat{v}_{ict}$ is the residual of the first-stage regression for household i in cluster c at time t and N_1 and N_0 are the number of observations in treated and control clusters respectively. Randomization inference constructs the distribution for the test statistic for every possible permutation of the randomization across clusters.¹⁸ In practice, given the large number of possible permutations (2,704,156 in our case), it is not possible to compute the test statistic for every possible permutation of the random allocation. We instead use 100,000 randomly selected permutations to construct the distribution. The p-value is then constructed based on the proportion of test statistic values that are greater than the actual test statistic value.

In each of the estimation tables, we report clustered standard errors computed using the cluster correlated Huber–White estimator, as well as the p-values of tests of the null that the coefficient is 0 computed using both the wild-bootstrap cluster-t procedure and randomization inference. Moreover, in Appendix C, we perform a Monte Carlo exercise where we compare the test size for these two approaches with the nominal test size, within data-generating processes that incorporate the main features of our data (number of clusters, number of observations and intra-cluster correlation). The simulations indicate that both inference methods perform relatively well.

4.2. Outcomes

In line with the theoretical model, our outcomes of interest span six domains: nutrition knowledge, child's consumption at under and over six months of age, household consumption, adult labor supply, and child's physical growth and morbidity. For child health and morbidity, which were the main focus of the intervention, we focus on children aged over 6 months, for whom the intervention would have completed. We pool data from the 2008–09 and 2009–10 follow-up surveys for the analysis. Details on the various measures within each domain are provided in Appendix D. However, two points are worth highlighting here. First, child consumption is measured from maternal reports of the foods consumed by each child. Second, special care was taken to measure household consumption rather than household expenditures. This is important in this context, since a large proportion of consumption is self-produced rather than purchased from a market.

Within each domain, we have several outcome measures, meaning that we end up with over 30 outcome variables. To limit the problem caused by multiple inference (the probability of rejecting a test is increasing in the number of tests carried out), we aggregate the multiple outcome measures within a domain into a summary index, following Anderson (2008).¹⁹ The index is a weighted mean of the standardized values of the outcome variables (with outcome variables re-defined so that higher values imply a better/more desirable outcome), with the weights calculated to maximize the amount of information captured in the index by giving less weight to outcomes that are highly correlated with each other. Another benefit of averaging across outcomes is that power is increased by reducing measurement error. Table E1 in

¹⁸ Randomization inference is non-parametric and exploits the randomization, rather than asymptotic results, for inference. A disadvantage, however, is that inference is conducted on a sharp null hypothesis of no effect for any unit in the data, rather than the more interesting hypothesis of null average effect.

¹⁹ While this helps to limit the problem of multiple inference, it does not address it fully because we still use eight indices. Indeed, if the data on the eight indices were independent, the familywise error rate would be at 40%. Adjusting for multiple inference within domains but not across domains is the most commonly used option (see for instance Finkelstein et al. (2012)).

¹⁶ For binary outcomes, results using probit models are very similar and are not reported.

¹⁷ Adult consumption also may be affected but, unfortunately, no good measure of adult-specific goods is available in our data.

Table 2
Effects on summary indices.

	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
	Main respondent's knowledge on nutrition	Child food consumption		Household food consumption	Labor supply		Child physical growth	Child morbidity (reversed)
		< 6 months	> 6 months		Adult males	Adult Females	> 6 months	> 6 months
T_c	0.167 ⁺	0.250*	0.143 ⁺	0.218*	0.262 ⁺	0.018	0.102*	−0.013
Standard error	[0.085]	[0.098]	[0.074]	[0.082]	[0.131]	[0.165]	[0.036]	[0.102]
Wild-cluster bootstrap-t p-value	{0.070}	{0.016}	{0.076}	{0.020}	{0.074}	{0.915}	{0.022}	{0.861}
Randomization inference p-value	{0.065}	{0.028}	{0.099}	{0.030}	{0.062}	{0.903}	{0.035}	{0.920}
Observations	1512	151	1280	3200	3642	4138	2175	2356
R-squared	0.098	0.214	0.099	0.063	0.183	0.136	0.026	0.053
Intra-cluster correlation	0.169	0.041	0.085	0.087	0.146	0.140	0.021	0.150
Mean control areas	−0.040	−0.109	−0.054	−0.099	−0.135	−0.050	0.266	0.022

Notes to table: Standard errors computed using the cluster-correlated Huber–White estimator are reported in square brackets, with clustering at the level of the cluster (at which treatment was assigned); wild-cluster bootstrap-t p-values and randomization inference p-values in curly brackets. All regressions include controls for cluster-level average education and Chewa ethnicity, both measured in 2004, and dummies for month of interview. All regressions other than in column 4 include controls for age and age-squared. Regressions in columns 2, 3, 7 and 8 also control for gender, while those in columns 5 and 6 control for education. Outcome variables are summary indices of variables relating to that domain of outcomes. They are constructed as described in Section 4.2. Higher values of the index in columns 7 and 8 indicate lower morbidity. The component variables for each index are outlined in Table E1 in Appendix E. Sample of children includes all those born after the intervention began in July 2005, and were therefore aged 0–53 months at time of interview. Specific samples are as follows. Column 1: all households present in waves 1 and 2 with a female main respondent aged 15 years or more; column 2: all children at wave 2 aged <6 months (some components of food consumption for this group not measured at wave 1); column 3: all children at wave 2 aged 6–53 months (food consumption for this group not measured at wave 1); column 4: all households at waves 1 or 2; columns 5 (6): all adult males (females) aged 15–65 years at waves 1 or 2; columns 7 and 8: all children at wave 1 or wave 2 aged 6–53 months. Note small discrepancies in samples between columns 7 and 8 due to missing values of outcome indicators.

* $p < 0.05$.** $p < 0.01$.+ $p < 0.1$.

Appendix E reports the outcomes used to compute the index associated with each domain.

By using a summary index, our results provide a statistical test for whether the intervention has a ‘general effect’ on each of the six main domains being tested that is robust to concerns about multiple inference (Kling et al., 2007; Liebman et al., 2004). However, because it is not possible to assess the magnitude of the effect from the results using the index, we also report the results on individual outcome variables.

Descriptive statistics pertaining to the outcomes and the indices for households and individuals in the control clusters are provided in Table E2. The table indicates that maternal knowledge on infant nutrition is mixed: questions related to weaning and nutritious value of foods were mostly correctly answered, while those related to food preparation and feeding when the child or its mother is unwell were often incorrectly answered. The food intake information indicates poor feeding practices: almost half of infants aged under 6 months were given water, while each of the protein-rich foods was consumed by fewer than half of children aged over 6 months. Low consumption of protein-rich foods is also apparent from the data on household consumption. Labor supply rates are similar for males and females: over 80% have at least one paid job, while around 9% have an additional job; both men and women work on average around 25 hours weekly. Finally, child health in this setting is very poor: the average child has a height-for-age z-score that is below −2 standard deviations of the WHO benchmark (and thus is considered to be stunted) and the incidence of illness is relatively high.

5. Results

We first show the impacts on all six composite indices, pooled across waves in Table 2 and separated by wave in Table 3. The subsequent tables (Tables 4–9) display the impacts on the sub-components of each index for those indices that show an overall statistically significant effect.²⁰ Note that, for ease of reading, each of Tables 4–9 reproduces, in its first column, the summary index from Table 2.

²⁰ Tables E3 and E4 display results for the sub-components of indices that do not show a statistically significant intervention impact.

5.1. Overall findings

Table 2 displays intervention impacts on all six composite indices, as described in Section 4.2. For child-level outcomes, we estimate the impacts on children born after the intervention began in July 2005, as these are the ones whose mothers were eligible to be visited by the peer counselor. This means that we consider impacts for children aged up to 4.5 years at the time of the second follow-up survey. Furthermore, since the intervention was ongoing at follow-up, we estimate impacts separately for children aged under 6 months (whose mothers were potentially being visited by the counselors at the time) and those aged over 6 months, and report impacts on health outcomes for the latter group only. For household and adult outcomes, we consider impacts on our entire sample, regardless of whether the household was directly exposed to the intervention and of the household's fertility choices.

The key rationale underlying the intervention is that households are inefficient producers of child health because they do not have the correct knowledge. In other words, the child health production function that households optimize over is ‘distorted’. In line with this, column 1 of Table 2 reports that the intervention improved main respondent's (mostly mothers) knowledge of child nutrition.²¹ The effect is only significant at the 10% level, possibly due to the high intra-cluster correlation in this variable. These improvements in knowledge translated into improved food consumption for both children aged under 6 months and children aged over 6 months (columns 2 and 3 in Table 2).^{22,23} The positive impact on the latter group imply that benefits of the intervention were retained even once the peer counselor stopped visiting the household.

Though the intervention provides no monetary or in-kind resources, household consumption could increase (see Section 3). In line with this, column 4 of Table 2 shows that the intervention increases total

²¹ The knowledge index was constructed from questions designed in consultation with program staff and tailored to the content of the intervention. Though the questions were piloted, no formal validation exercise was conducted.

²² Note that child-specific consumption for children over 6 months is measured at second follow-up only.

²³ That the intervention improved both knowledge and child nutrition suggests that improving knowledge of the child health production function improves nutrition choices. One might want to test this mechanism directly using the intervention as an instrument for knowledge. Unfortunately, the intervention impact on knowledge is not sufficiently strong to allow us to do this without encountering a weak instrument problem.

Table 3
Effects on summary indices by wave.

	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
Top panel: wave 1 results								
Main respondent's knowledge on nutrition		Child food consumption		Household food consumption	Labor supply		Child physical growth	Child morbidity (reversed)
		< 6 months	> 6 months		Adult males	Adult females	> 6 months	> 6 months
T_c	0.195			0.156	0.183	−0.016	0.093	0.027
Standard error	[0.136]			[0.113]	[0.135]	[0.163]	[0.045]	[0.103]
Wild-cluster bootstrap-t p-value	{0.228}			{0.212}	{0.216}	{0.985}	{0.108}	{0.769}
Randomization inference p-value	{0.143}			{0.206}	{0.244}	{0.920}	{0.107}	{0.853}
Observations	1512			1644	1790	2080	932	1061
R-squared	0.079			0.069	0.177	0.157	0.032	0.040
Intra-cluster correlation	0.156			0.141	0.140	0.183	0.026	0.175
Mean control areas	−0.054			−0.075	−0.119	−0.033	0.286	0.001
Bottom panel: wave 2 results								
Main respondent's knowledge on nutrition		Child food consumption		Household food consumption	Labor supply		Child physical growth	Child morbidity (reversed)
		< 6 months	> 6 months		Adult Males	Adult Females	> 6 months	> 6 months
T_c	0.152	0.250*	0.143 ⁺	0.305**	0.323*	0.050	0.112*	−0.051
Standard error	[0.119]	[0.098]	[0.074]	[0.092]	[0.148]	[0.193]	[0.040]	[0.124]
Wild-cluster bootstrap-t p-value	{0.273}	{0.016}	{0.076}	{0.002}	{0.036}	{0.877}	{0.022}	{0.743}
Randomization inference p-value	{0.248}	{0.028}	{0.099}	{0.014}	{0.036}	{0.768}	{0.032}	{0.746}
Observations	1512	151	1280	1556	1852	2058	1243	1295
R-squared	0.043	0.214	0.099	0.050	0.184	0.125	0.028	0.043
Intra-cluster correlation	0.190	0.041	0.085	0.085	0.192	0.150	0.017	0.197
Mean control areas	−0.035	−0.109	−0.0541	−0.132	−0.148	−0.073	0.238	0.045

Notes to table: Standard errors computed using the cluster-correlated Huber–White estimator are reported in square brackets, with clustering at the level of the cluster (at which treatment was assigned); wild-cluster bootstrap-t p-values and randomization inference p-values in curly brackets. All regressions include controls for cluster-level average education and Chewa ethnicity, both measured in 2004, and dummies for month of interview. All regressions other than in column 4 include controls for age and age-squared. Those in columns 5 and 6 also control for individual education. Finally, those in columns 2, 3, 7 and 8 also control for gender. Outcome variables are summary indices of variables relating to that domain of outcomes. They are constructed as described in Section 4.2. Higher values of the index in column 8 indicate lower morbidity. The component variables for each index are outlined in Table E1 in Appendix E. Sample of children includes all those born after the intervention began in July 2005, and were therefore aged 0–53 months at time of interview. Specific samples are as follows. Column 1, both panels: all households present in both waves 1 and 2 with a female main respondent aged 15 years or more; column 2, bottom panel: all children at wave 2 aged under 6 months; column 3, bottom panel: all children at wave 2 aged 6–53 months (food consumption for this group not measured at wave 1); column 4, top (bottom) panel: all households at wave 1 (2); column 5, top (bottom) panel: all adult males aged 15–65 years at wave 1 (2); column 6, top (bottom) panel: all adult females aged 15–65 years at wave 1 (2); columns 7 and 8, top (bottom) panel: all children at wave 1 (2) aged 6–44 months (6–53 months). Note small discrepancies in samples between columns 7 and 8 due to missing values of outcome indicators. Knowledge index in wave 1 constructed with three questions asked in wave 1; and that in wave 2 with four questions asked in wave 2 only.

* $p < 0.05$.

** $p < 0.01$.

⁺ $p < 0.01$.

household food consumption, measured using the composite index, at 5% significance. The increase in household consumption might have been partially funded by improvements in adult labor supply, particularly of males (column 5); female labor supply is unchanged by the intervention (column 6). Although our model of Section 3 already indicated that labor supply could increase, other factors may also explain increased consumption, including borrowing and/or drawing down savings. Increases in labor supply could also be due to a reduction in time devoted to caring for sick children.

A key policy question is whether the observed adjustments on various margins of household behavior (increased consumption and labor supply) improved child health. Column 7 shows that these changes in behavior translate into improved child physical growth for children aged over 6 months. No significant effect is found on child morbidity.²⁴ Note though

that given the substantial infant mortality reductions found by Lewycka et al. (2013), and under the assumption that weaker children are the ones more likely to survive as a result of the intervention (Bozzoli et al., 2009; Deaton, 2007), the reported effects likely underestimate the true effect of the intervention on child health.

Table 3 shows the results by follow-up survey round ('wave'), which are of interest in order to see whether the effects are sustained over time. In general, the table shows that the point estimates share the same signs across both waves and are not significantly different from each other. Notably, the point estimates of household food consumption, male labor supply and child physical growth all show a tendency to be larger in wave 2 than in wave 1, and they are statistically significant in wave 2 only, although they are not significantly different from the wave 1 estimates.²⁵ The tendency for larger treatment effects on consumption and male labor supply in wave 2 may be due to some heterogeneity of treatment effect according to the time when the surveys

²⁴ We also considered the intervention impacts on child anthropometrics and morbidity for children under 6 months who were undergoing the intervention at the time of the survey and for whom these would be intermediate-stage data. We find a positive but statistically insignificant effect on both outcomes. Interestingly, we find that the prevalence of diarrhea decreases for children under 6 months, consistent with the reduced intake of water and non-maternal milk for this group.

²⁵ Note that there are more children aged over 6 months who would have been eligible for the intervention in wave 2 than in wave 1 since the former includes children born between July 2005 and July 2009 while the latter includes children born between July 2005 and October 2008.

Table 4
Effects on components of the knowledge index.

	[1] Summary index	[2] Breastfeeding when infant has diarrhea	[3] Are biscuits or groundnuts/soya more nutritious for children aged 6 months to 3 years?	[4] From what age should solid foods be given to infants?	[5] How should an HIV-positive woman feed her baby?	[6] Is nsima or porridge more nutritious for an infant aged >6 months?	[7] What is the best way to cook fish with porridge for an infant aged >6 months?	[8] Should eggs be given to an infant aged >9 months?
T_c	0.167 ⁺	0.253 ⁺	−0.052	0.037	0.138	−0.101	0.067 ^{**}	0.104
Standard error	[0.085]	[0.115]	[0.041]	[0.026]	[0.150]	[0.078]	[0.019]	[0.069]
Wild-cluster bootstrap-t p-value	{0.070}	{0.084}	{0.290}	{0.166}	{0.444}	{0.210}	{0.002}	{0.186}
Randomization inference p-value	{0.065}	{0.028}	{0.222}	{0.292}	{0.399}	{0.179}	{0.008}	{0.192}
Observations	1512	1512	1512	1512	1512	1512	1512	1512
R-squared	0.098	0.104	0.052	0.043	0.045	0.072	0.038	0.020
Intra-cluster correlation	0.169	0.277	0.082	0.049	0.408	0.183	0.057	0.107
Mean control areas	−0.040	0.217	0.938	0.88	0.393	0.857	0.026	0.719

Notes to table: All regressions include controls for age, age-squared, average cluster-level education and Chewa ethnicity, both measured in 2004, and dummies for month of interview. Standard errors computed using the cluster-correlated Huber–White estimator are reported in square brackets, with clustering at the level of the cluster (at which treatment was assigned); wild-cluster bootstrap-t and randomization inference p-values in curly brackets. Sample includes all households with a female main respondent aged 15 years and over. ‘Summary Index’ aggregates the measures in columns 2–8 using the method described in Section 4.2. The variables in columns 2–8 are dummy variables equal to 1 if the respondent answered correctly. Questions in columns 2–6 and column 8 were multiple-choice questions where respondents chose one correct answer from three–five options. Question in column 7 was an open-ended question, with interviewers marking correctly answered options.

* $p < 0.05$.

** $p < 0.01$.

+ $p < 0.1$.

were conducted. Wave 1 data were collected between mid November and the end of March, while wave 2 data were collected between October and early January. The levels of the consumption and male labor supply indices are the lowest in the October to mid November period, which is when the treatment effect is the highest.

While the composite indices allow us to assess the general impact of the intervention on each domain, their magnitudes cannot be interpreted, as the weighting used to build the index distorts the scale. To shed more light on the magnitude of the effects, we next report and discuss findings for individual outcomes for the composite indices for which there is a statistically significant effect of the intervention. We note that the results on the index components must be considered

Table 5
Effects on intake of liquids by children under 6 months.

	[1] Summary index	[2] Water	[3] Milk other than maternal
T_c	0.250 [*]	−0.107	−0.082 [*]
Standard error	[0.098]	[0.069]	[0.034]
Wild-cluster bootstrap-t p-value	{0.016}	{0.122}	{0.012}
Randomization inference p-value	{0.028}	{0.212}	{0.115}
Observations	151	151	151
R-squared	0.214	0.362	0.087
Intra-cluster correlation	0.041	0.000	0.060
Mean control areas	−0.109	0.474	0.101

Notes to table: All regressions include controls for age, age-squared, gender, average cluster-level education and Chewa ethnicity, both measured in 2004, and dummies for month of interview. Standard errors computed using the cluster-correlated Huber–White estimator are reported in square brackets, with clustering at the level of the cluster (at which treatment was assigned); wild-cluster bootstrap-t and randomization inference p-values in curly brackets. Sample includes children at wave 2 aged under 6 months. ‘Summary index’ aggregates the measures in columns 2–3 using the method described in Section 4.2. ‘Water’ is an indicator for whether the child had any water in the three days prior to the survey. ‘Milk other than maternal’ is an indicator (measured in second follow-up only) for whether the child had milk other than breastmilk in the three days prior to the survey.

* $p < 0.05$.

** $p < 0.01$.

+ $p < 0.1$.

exploratory and interpreted carefully since the familywise error rate is not being controlled for. Appendix F reports results on individual outcomes by wave.

5.2. Nutritional knowledge, consumption and labor supply

The intervention resulted in improvements in the main respondent’s knowledge of child nutrition. The index aggregates together the correct responses to seven questions (reproduced in Appendix G). Columns 2–8 of Table 4 report the impact of the intervention in terms of the proportion of respondents who correctly answered each of the seven questions. The results show that the knowledge improvements are concentrated on breastfeeding practices when infants are ill and on knowledge of food preparation practices. We note that the intra-cluster correlation coefficient is very high for most components of the index, which makes it particularly difficult to detect statistically significant differences.²⁶

Improvements in food consumption were detected for children under and over 6 months. For the former group, we see from Table 5 that the improvement comes from a reduction in non-maternal milk. There is also a reduction (though not statistically significant) in the consumption of water. Table 6 shows that improvements for the latter group are driven by substantially higher consumption of protein-rich beans in the three days prior to the interview. The intakes of meat and eggs (also protein rich) are also positive, although not statistically significant, most likely due to the reduced sample size (child food intake was collected at second follow-up only). Overall, these results indicate that the intervention significantly affected the composition of child nutritional intake.

²⁶ Note that the number of observations is lower than for other household-level variables. This is because we combine wave 1 and wave 2 questions into a single index, to maximize its informational content, and drop households without a female main respondent aged 15 years or above. Note that the three questions in wave 1 are a subset of the seven questions asked in wave 2. We construct the index to include responses from wave 1 to the three common questions and responses to the four questions unique to wave 2. This is because there was evidence of households having learnt or found out answers to the three questions carried over from wave 1 to wave 2.

Table 6
Effects on food consumption by children aged 6–53 months.

	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
	Summary index	Any beans	Any meat	Any fish	Any eggs	Any Vegetables	Any fruit	Any nsima	Any porridge
T_c	0.143 ⁺	0.225**	0.089	0.006	0.025	−0.010	−0.009	0.025	0.096
Standard error	[0.074]	[0.056]	[0.095]	[0.099]	[0.052]	[0.020]	[0.058]	[0.015]	[0.064]
Wild-cluster bootstrap-t p-value	{0.076}	{0.002}	{0.474}	{0.925}	{0.655}	{0.723}	{0.941}	{0.144}	{0.208}
Randomization inference p-value	{0.099}	{0.007}	{0.289}	{0.954}	{0.632}	{0.634}	{0.895}	{0.140}	{0.251}
Observations	1280	1280	1280	1280	1280	1280	1280	1280	1280
R-squared	0.099	0.067	0.021	0.012	0.010	0.142	0.153	0.144	0.035
Intra-cluster correlation	0.085	0.116	0.084	0.112	0.048	0.018	0.093	0.000	0.136
Mean control areas	−0.054	0.258	0.290	0.462	0.163	0.959	0.699	0.929	0.800

Notes to table: All regressions include controls for age, age-squared, gender, average cluster-level Chewa ethnicity and education, both measured in 2004, and dummies for month of interview. Standard errors computed using the cluster-correlated Huber–White estimator are reported in square brackets, with clustering at the level of the cluster (at which treatment was assigned); wild-cluster bootstrap-t and randomization inference p-values in curly brackets. Sample contains all children at wave 2 aged 6–53 months (data on child solid intake collected at wave 2 only). ‘Summary index’ aggregates the measures in columns 2–9 using the method described in Section 4.2. The variables in columns 2–9 are dummy variables equal to 1 if the corresponding food was consumed by the child in the three days prior to the survey.

* $p < 0.05$.

** $p < 0.01$.

⁺ $p < 0.1$.

Table 7
Effects on household food consumption.

	[1]	[2]	[3]	[4]	[5]
	Summary index	Per-capita monthly food consumption for:			
	Summary index	Cereals	Proteins	Fruit and vegetables	Other foods
T_c	0.218*	−9.768	129.150 ⁺	269.987*	60.701
Standard error	[0.082]	[52.432]	[54.802]	[108.591]	[33.552]
Wild-cluster bootstrap-t p-value	{0.020}	{0.863}	{0.066}	{0.044}	{0.126}
Randomization inference p-value	{0.030}	{0.865}	{0.025}	{0.033}	{0.069}
Observations	3200	3200	3200	3200	3200
R-squared	0.063	0.117	0.020	0.195	0.025
Intra-cluster correlation	0.087	0.074	0.042	0.173	0.053
Mean control areas	−0.099	605.80	349.10	679.80	149.50

Notes to table: Standard errors computed using the cluster-correlated Huber–White estimator are reported in square brackets, with clustering at the level of the cluster (at which treatment was assigned); wild-cluster bootstrap-t and randomization inference p-values in curly brackets. Sample includes all households at waves 1 or 2. All regressions include controls for age, age-squared, average cluster-level education and Chewa ethnicity, both measured in 2004, and dummies for month of interview. Coefficients in columns 2–5 are in terms of Malawi Kwacha. (The average exchange rate to the US Dollar was approximately 140MK = 1 US\$ at the time of the surveys.) ‘Summary index’ is an index of the food items in columns 2–5, constructed as described in Section 4.2. ‘Cereals’ includes consumption of rice, maize flour and bread. ‘Proteins’ includes consumption of milk, eggs, meat, fish and pulses. ‘Fruit and vegetables’ includes consumption of green maize, cassava, green leaves, tomatoes, onions, pumpkins, potatoes, bananas, masuku, mango, ground nuts and other fruits and vegetables. ‘Other foods’ includes consumption of cooking oil, sugar, salt, alcohol and other foods.

* $p < 0.05$.

** $p < 0.01$.

⁺ $p < 0.1$.

We saw from Table 2 that the intervention resulted in improvements in overall household food consumption. Columns 2–5 of Table 7 show that the improvement is due to an increase in the consumption of proteins and of fruit and vegetables. The effects are relatively large. Focusing on proteins, which are particularly important for child growth as shown by, for example, Puentes et al. (2014), we decompose the effect on the extensive (i.e. moving from consuming no proteins to some proteins) and intensive margin (i.e. moving from consuming some proteins to more proteins).²⁷ Around 26% of households in control clusters report consuming no protein-rich foods in the seven days prior to interview; hence there is clear potential for improvement in the extensive margin. Indeed, the extensive margin accounts for one-third of the consumption increase.²⁸ The increase in the intensive margin

corresponds to 210 g of meat/poultry extra and 640 g of beans extra per child per month. To put these quantities in perspective, a toddler will usually consume 50 g of beans in one portion, together with some vegetables and carbohydrates.

A number of factors are likely to explain this substantial increase in food consumption: first, the time span of the intervention is sufficiently long (it had already been up and running for over 3.5 years by the time consumption was first measured); second, the intervention was intensive, involving up to five one-to-one home visits; third, as seen from the labor supply results in Table 2, there was scope for labor supply to increase and thereby fund at least some of the increased consumption.

Table 2 also showed that the male labor supply index increased as a result of the intervention. Looking at the sub-components of the index – work participation, likelihood of having at least two jobs and hours worked – Table 8 reports positive effects of the intervention on all three, though only statistically significant for the probability of having at least two jobs. However, the intra-cluster correlations are much higher for work participation and for the number of hours worked than for the probability of having at least two jobs (0.213 and 0.100 vs. 0.033), which greatly reduces the power to detect a significant effect of the intervention on the former.

²⁷ Calculations available upon request.

²⁸ The consumption increase coming from the extensive margin is calculated under the assumption that the households in the treated clusters induced to consume protein-rich foods as a result of the intervention all consume proteins equivalent to the average consumed by control cluster households with non-zero protein consumption. The increase on the intensive margin – corresponding to the rest of the consumption increase – is further decomposed into food quantities (beans and meat/poultry) under the assumptions that the entire amount is consumed by children aged under 12 years only (who are, in control clusters, 2.4 per household on average) and that households pay prices equivalent to the average cluster-level median unit values.

Table 8
Effects on male labor supply.

	[1]	[2]	[3]	[4]
	Summary index	Works	Has at least 2 jobs	Weekly hours worked
T_c	0.262 ⁺	0.106	0.080 ^{**}	4.310
Standard error	[0.131]	[0.080]	[0.025]	[2.918]
Wild-cluster bootstrap-t p-value	{0.074}	{0.272}	{0.010}	{0.240}
Randomization inference p-value	{0.062}	{0.220}	{0.011}	{0.202}
Observations	3642	3642	3642	3642
R-squared	0.183	0.183	0.057	0.163
Intra-cluster correlation	0.146	0.213	0.033	0.100
Mean control areas	−0.135	0.819	0.094	25.740

Notes to table: All regressions include controls for age, age-squared, average cluster-level education and Chewa ethnicity, both measured in 2004, individual education and dummies for month of interview. Standard errors computed using the cluster-correlated Huber–White estimator are reported in square brackets, with clustering at the level of the cluster (at which treatment was assigned); wild-cluster bootstrap-t and randomization inference p-values in curly brackets. Sample includes all males aged 15–65 years at wave 1 or 2. ‘Summary index’ contains the variables in columns 2–4 and is computed using the method described in Section 4.2. ‘Works’ is an indicator of whether individual had an income-generating activity at the time of the survey. ‘Has at least 2 jobs’ is an indicator of whether individual has at least two income-generating activities. ‘Weekly hours worked’ gives the total hours worked in the week prior to the survey on both income-generating activities.

* $p < 0.05$.

** $p < 0.01$.

+ $p < 0.1$.

The finding that the intervention increases male labor supply is consistent with it being a margin with considerable scope for increase. Indeed, previous research in Malawi has shown that labor supply is upward sloping rather than fixed (Dimova et al., 2010; Goldberg, 2016). In our data, only 9% of males in control clusters have a second job, most of them in non-agricultural self-employment activities.²⁹ Moreover, there is considerable entry into and exit from secondary jobs: among those with (without) a secondary job at first follow-up, 33% (7%) have one at second follow-up, a year later. While an extensive literature has documented increases in labor supply in response to increases in uncertainty and income shocks in developing countries (Ito and Takashi, 2009; Kochar, 1999; Lamb, 2003; Rose, 2001; Saha, 1994), this is the first paper to document labor supply responses to changes in the perceived child health production function.

Beyond the mechanism for the increase in labor supply indicated in Section 3, important cultural features of Malawian society are likely to contribute to the increase in male, rather than female, labor supply. In particular, the main ethnic group in Mchinji – the Chewa – is a matrilineal and matrilineal group, where men usually move to their wives’ villages on marriage, and wealth (predominantly land) is held by women and passed on through the matriline (Phiri, 1983; Sear, 2008). As a consequence, women have more power and authority than in patrilineal societies common across most of Africa and South Asia (Reniers, 2003). Indicative of this empowerment, all three measures of labor supply are strikingly similar for males and females (last row of Table 8 and Table E3).³⁰ Finally, mothers are generally the main caregivers of children. So the finding that male labor supply increases in response to the mother receiving information on child nutrition is in line with the cultural background, where females are relatively empowered.

²⁹ Over half of these second jobs involve employment in own/family business, a quarter involve work on the family farm and the rest involve work as an employee in public/private sector (~20%) or on someone else’s farm (<5%).

³⁰ This has been documented by others for the Malawian context, including Goldberg (2016) and the 2004 Malawi DHS Report (National Statistical Office, Malawi and ORC Macro, 2005, pp. 34–36). In the matrilineal Khasi society (India), women and men also have similar labor supply profiles (Gneezy et al., 2009).

Table 9
Effects on physical growth of children aged 6–53 months.

	[1]	[2]	[3]	[4]
	Summary index	Height for age	Healthy weight for age	Healthy weight for height
T_c	0.102 [*]	0.274 [*]	0.028	0.042 ⁺
Standard error	[0.036]	[0.100]	[0.017]	[0.024]
Wild-cluster bootstrap-t p-value	{0.022}	{0.022}	{0.120}	{0.132}
Randomization inference p-value	{0.035}	{0.055}	{0.308}	{0.147}
Observations	2175	2175	2175	2175
R-squared	0.026	0.048	0.020	0.029
Intra-cluster correlation	0.021	0.023	0.018	0.014
Mean control areas	0.266	−2.326	0.829	0.852

Notes to Table: Standard errors computed using the cluster-correlated Huber–White estimator are reported in square brackets, with clustering at the level of the cluster (at which treatment was assigned); wild-cluster bootstrap-t and randomization inference p-values in curly brackets. All regressions include controls for age, age-squared, gender, dummies for month of interview and average cluster-level education and Chewa ethnicity, both measured in 2004. Sample includes children aged 6–53 months at wave 1 or 2. ‘Summary index’ contains the variables in columns 2–4 and is computed using the method described in Section 4.2. ‘Height for age’ is a standardized z-score relative to the WHO reference population. ‘Healthy weight for age’ is a dummy variable = 1 if child’s weight-for-age z-score is not more than 2 standard deviations above or below the WHO reference population. ‘Healthy weight for height’ is a dummy variable = 1 if child’s weight-for-height z-score is within 2 standard deviations of the WHO reference population.

* $p < 0.05$.

** $p < 0.01$.

+ $p < 0.1$.

5.3. Child health

Table 2 documented improvements in child physical growth for children over 6 months. Looking at the sub-components of the physical growth index in Table 9, we see that the improvements are due to an increase in the average height-for-age z-score by 0.27 of a standard deviation of the WHO norm.³¹ This is an important increase, and corresponds in magnitude to 65% of the average effect size obtained with the direct provision of food in food-insecure populations (Bhutta et al., 2008). Interestingly, further analysis, documented in Table E5, indicates that the effects on physical growth are much stronger for children aged 6–24 months.³²

Clearly, we cannot disentangle whether the improvement in physical growth is due to the reduction in intake of liquids other than breast milk when the child was under 6 months old or to the improvement in child food intake after age 6 months, or a combination of both. Our key message is that households responded to the intervention by increasing consumption and working more, which is the first such finding in this literature.³³

6. Alternative explanations

We have argued, using the model of Section 3, that consumption and labor supply will increase because the perceived productivity of child consumption (in terms of child health) increased as a result of the intervention. Here we consider four alternative explanations. First, we consider and rule out that the increases in adult labor supply are driven

³¹ As is common with anthropometric data from developing countries, the SD of the height-for-age z-score in our sample is larger than that in the WHO reference population (1.5 instead of 1), and so this increase corresponds to an increase of 0.18 of a SD using the SD for our sample.

³² These patterns are consistent with two non-competing explanations: that the intervention did not work very well at the beginning and/or children in control clusters experienced catch-up growth at slightly older ages.

³³ We have also examined the heterogeneity of the effect of the intervention on the anthropometric and morbidity indices according to whether the mother has had more than one child since the intervention started. The interaction terms were far from statistically significant (p-value of 0.45 or larger).

by improvements in adult health somehow generated by the intervention (Table E6). Second, parental investment in child nutrition could have increased as a result of decreased fertility caused by the intervention, potentially yielding an increase in child quality (Becker and Tomes, 1976). The intervention could have reduced fertility by reducing infant mortality and consequently inducing households to demand fewer children, or through the family planning component of the intervention. Analysis of the intervention effects on family planning behavior and births to women in our sample (as reported in the MaiMwana Health Surveillance System³⁴) reveals very small and statistically insignificant effects, ruling out this channel (Table E7).³⁵

Third, the reduction in infant mortality and improvement in child health could have affected parental labor supply, through changing the demand for childcare. It is plausible that if infant mortality declines and there are more surviving children, mothers in treated clusters may increase their time devoted to childcare, therefore working less, leading to fathers working more to compensate for this. However, as we showed in Section 5.1, the intervention does not appear to have reduced female labor supply, suggesting that this mechanism is not at play in our context. Another potential channel through which labor supply may change as a result of improvements in children's health is through reducing the need for fathers to be at home to help take care of children, thus facilitating an increase in their labor supply.

Finally, effects could also be driven by information provided by the intervention on issues other than infant feeding practices, e.g. vaccination of infants, promotion of HIV testing and hygiene practices. Though these could have improved child health, it is unlikely that they would improve household consumption and labor supply. Available evidence suggests that these other components would have had very modest or no effects. Lewycka et al. (2013) find mixed intervention effects on vaccination rates (BCG vaccination rates increased, while polio vaccination rates decreased). Moreover, vaccination rates in control clusters were high, leading to small intervention effects. The authors also find that the intervention was not effective in improving antenatal HIV counseling and treatment. This is not surprising, since the intervention simply encouraged women to get tested for HIV, without any efforts to alleviate cost constraints or stigma effects related to being tested (Derksen et al., 2014; Ngatia, 2011; Thornton, 2008). Finally, our finding that the intervention did not reduce the prevalence of diarrhea for children aged between 6 and 53 months and adults (Tables E4 and E6) suggests that the component on hygiene information probably had limited success.

7. Conclusion

In this paper, we use exogenous variation in mothers' knowledge of the child health production function induced by a cluster randomized intervention in Malawi, to study empirically whether improving knowledge of the child health production function influences a broad range of household behaviors.

We first document that the intervention improved mothers' knowledge of nutrition. Using a simple theoretical model, we show that households should react to this improved knowledge by changing the composition of child food intake in favor of protein-rich foods, fruit and vegetables. The intervention could also increase household food consumption and adult labor supply, although the theoretical

predictions are ultimately ambiguous. Our empirical results show that, indeed, both child's food intake and child nutritional status improved and that ultimately both labor supply and household food consumption increased.

We hypothesize that two issues might have contributed to the success of the intervention. First, the provision of information was not merely a one-off event in the intervention areas, but a sustained activity, still in place, that serves to spread information and to remind households of the importance of child nutrition on an ongoing basis. This may also explain why households adjusted on non-health margins to adhere to advice provided by this nutrition intervention and may shed light on why some health information campaigns have been successful while others have failed. Second, the main ethnic group in rural Malawi, the Chewa, is a matrilineal one, in which women are likely to have more bargaining power and authority within the household than women in patrilineal societies common in much of the rest of Africa and South Asia. This higher female empowerment might indicate that women are in a good position to implement the recommendations given by the counselors as well as to encourage fathers to work more. It is not clear whether such responses may emerge in other settings and we see this as an area worthy of further investigation.

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Appendix A. Supplementary Appendices and Data

Supplementary data to this article can be found online at <http://dx.doi.org/10.1016/j.jdeveco.2016.05.002>.

References

- Alderman, H., 2007. Improving nutrition through community growth promotion: longitudinal study of the nutrition and early childhood development program in Uganda. *World Dev.* 35 (8), 1376–1389.
- Alderman, H., Behrman, J., Lavy, V., Menon, R., 2001. Child health and school enrolment. A longitudinal analysis. *J. Hum. Resour.* 36 (1), 185–205.
- Almond, D., Currie, J., 2011. Human capital development before age five. In: Ashenfelter, O., Card, D. (Eds.), *Handbook of Labor Economics*, Volume 4, Part B: 1315–1486. Elsevier.
- Anderson, M.L., 2008. Multiple inference and gender differences in the effects of early intervention: a reevaluation of the Abecedarian, Perry Preschool, and early training projects. *J. Am. Stat. Assoc.* 103 (484), 1481–1495.
- Banerjee, A., Duflo, E., Glennerster, R., 2008. Putting a Band-Aid on a corpse: incentives for nurses in the Indian public health care system. *J. Eur. Econ. Assoc.* 5 (2–3), 487–500.
- Banerjee, A., Duflo, E., Postel-Vinay, G., Watts, T., 2010. long-run health impacts of income shocks: wine and phylloxera in nineteenth-century France. *Rev. Econ. Stat.* 92 (4), 714–728.
- Barham, T., 2012. Enhancing cognitive functioning: medium-term effects of a health and family planning program in Matlab. *Am. Econ. J. Appl. Econ.* 4 (1), 245–273.
- Becker, G., 1965. A theory of the allocation of time. *Econ. J.* 75 (299), 493–517.
- Becker, G., Tomes, N., 1976. Child endowments, and the quantity and quality of children. *J. Polit. Econ.* 84 (4), S143–S162.

³⁴ The MaiMwana Health Surveillance System interviews the mothers of all children born in the 24 clusters since 2005 at 1 month and 7 months of age, and thus provides a more complete picture of births in the study areas than cross-sectional surveys.

³⁵ Because the intervention decreased infant mortality, an alternative explanation for our findings is that (a) the children who survive tend to have worse health and (b) parents compensate for the worse health by providing them with more resources. Based on the results of Lewycka et al. (2013), we estimate that the marginal surviving children would be approximately 2.3% of the intervention sample, which is too small to explain the magnitude of the treatment effects if they were to be driven entirely by these marginal children.